

Learning Grasp-Stability with Multi-Finger-Hands and Touch in Simulation

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Abstract

Grasping objects has been a fundamental challenge in the field of robotics for many years. A successful grasp requires a careful balancing of contact forces as well as an optimal grasp position at the object. Due to this complex requirement, it is hard to use only indirect measurement, such as vision, to determine if a given grasp will be successful or not. A new promising approach to overcome this difficulty is the use of direct measurements like tactile sensing to determine contacts directly. To analyze the value of touch sensing for predicting grasp outcomes, it is crucial to understand how touch sensing influences the prediction accuracy when combined with other modalities, especially vision.

This work investigates whether the use of touch sensing aids in predicting grasp outcomes of a multi-finger hand in simulation. Therefore, more than 56,000 grasping trials were collected using a four-finger hand equipped with omnidirectional high-resolution optical tactile sensors at the fingertips. The collected grasp data was used to train and evaluate an end-to-end visuo-tactile deep neural network to predict the outcome of a given grasp from either one single-modality, and from vision and touch in combination.

The results of this work indicate that the additional use of tactile readings aids in predicting grasp outcomes of a multi-finger hand in simulation compared to purely visual approaches.

Keywords: Robot Grasping, Tactile Sensing, Robot Learning, Machine Learning, Simulation

Statement of authorship

I hereby certify that I have authored this document entitled *Learning Grasp-Stability with Multi-Finger-Hands and Touch in Simulation* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. During the preparation of this document I was only supported by the following persons:

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1 Introduction

Robotics promises to be one of the most disruptive technologies in the annals of human progress. The profound impact of an affordable and potentially unlimited work capacity could have a lasting effect on our daily lives and, by extension, our social fabric. Yet, the widespread use of robots in our society is still a long way off, apart from certain specialized areas of industry. The mastery of object manipulation by robots is of great importance in meeting the challenge posed by the widespread use of robots in our society. This capability is crucial as it allows robots to seamlessly interact and navigate with the physical world, mimicking the dexterity and precision of humans. When robots are able to manipulate objects effectively, they become versatile tools that can perform a wide range of tasks in different domains. Acquiring the ability to grasp objects is a crucial first step towards mastering the broader field of object manipulation, as it forms the basis for robots' interaction with the physical world. By enabling a robot to precisely grasp any object, it opens up a wide range of new application areas from industry to care to medicine.

1.1 Motivation

For a single grasp with a multi-finger hand, a human makes use of different perception factors, including vision and tactile sensing [20]. While the visual input is enough for a fast object localization and a first grasp position planning, only the direct tactile feedback provides humans with the ability to control contact forces and find a final hand position. This ability leads to successful grasps, even if the visual perception of the object is only possible to a limited extent [18]. Whereas in the field of robotics, for many years only visual feedback was used to plan grasping, some recent work demonstrates the use of tactile sensing for achieving better results in grasping objects [8], [7].

Although the use of tactile sensors seems to be a beneficial approach, they are slow of getting involved into robotics and especially robot grasping. Integrating tactile sensing into grasp planning can be challenging due to hardware limitations of the technologies employed. Additionally, tactile readings are difficult to model, such that even small errors in measurement and calibration can cause significant performance reductions of analytical models.

A promising approach for building analytical models that make use of tactile readings is Machine Learning (ML). Some recent works proposed vision-based grasping approaches relying on end-to-end training to directly predict grasp outcome or grasp position based on camera images [28], [29]. However, the importance of tactile sensing for humans raises the question whether the additional use of tactile readings for training end-to-end models such as deep neural networks will lead to more accurate grasp predictions?

To answer this question, it is crucial to take a closer look at the design of a multi-modal deep learning model for predicting grasp stability. Due to the possibility of predicting a grasp success probability, we are able to adjust the grasp position of the hand to obtain a more promising and stable grasp position. Especially in critical situations like grasping and lifting fragile objects, this feature is a further step in robot grasping, since the grasp configuration can be adjusted until a defined success probability threshold is reached. By examining models trained on different modalities, one is able to give an answer to whether the use of tactile readings will help in predicting a more accurate grasp stability in comparison to purely vision based methods.

Considering the fact that experiments with robots in the real world are often expensive in money and time, the execution of grasping trials is done in a simulation. This not only makes it easier to perform experiments with robots from anywhere at low cost, but also enables new dimensions of data mining in a fraction of the time it would take with real robots. As a result, both the tactile and visual measurements are retrieved by the simulation, necessitating a closer look at whether the retrieved data are of sufficient quality to successfully train the models. This also raises the question if the use of tactile sensors in simulation is capable of overcoming hardware limitations of this tactile sensors in real world experiments.

1.2 Contributions

This study seeks to investigate the extent to which the incorporation of tactile sensing contributes to the prediction of grasp outcomes in the context of a multi-fingered robotic hand operating within a simulated environment.

Therefore, this work proposes end-to-end deep learning models trained on either vision and touch as well as an end-to-end multi-modal model incorporating vision and touch to predict the outcome of a given grasp. The study of the model performance allows evaluating the relative importance of each single modality as well as the multi-modal approach.

To generate the necessary training data, a methodology involving parallel grasp executions within a high-fidelity simulation environment is introduced. The tactile information is sourced from omnidirectional high-resolution tactile sensors, which provide deformation images upon establishing contact with the grasp object. Utilizing this dataset, distinct models are trained based on vision-only, tactile-only, and vision-and-tactile modalities to make grasp outcome predictions.

This work is the first proposing end-to-end single and multi-modal models for predicting the grasp outcome of a multi-finger in simulation and provides a detailed evaluation of each modality for predicting grasp success. The chosen approach stands out compared to other approaches since a physical robot is not necessary, which causes a massive improvement in time, money, and number of conducted experiments. Additionally, the proposed grasping strategy is much simpler compared to manually designed grasping metrics. Nevertheless, the results show that the additional use of tactile sensing improves overall grasping performance.

1.3 Outline

The thesis commences with an extensive review of relevant literature in Chapter 2, encompassing topics central to the thesis. Following this, Chapter 3 provides an introductory framework outlining the problem statement and the formal context within which the research is situated. Chapter 4 dives into the kinematics of grasping in simulation. The complete experimental grasp cycle, as well as the collection of data, is described and evaluated. All data used for training is obtained in the simulation, which makes it necessary to first discuss the simulation to better interpret model architectures and performance values of the models. To create a connection of model performance and chosen architecture, Chapter 5 provides an overview of the

model architecture proposed in this work. Particular attention is paid to the requirements for deep neural networks based on the number and type of input, both for the single-modal and multi-modal models. Chapter 6 explains the experimental setup and shows the results pointing to the help of touch sensing for more accurate grasp prediction. To allow a comprehensive analysis of the results, additional attention is drawn to the training behavior as well as the confusion statistics of the proposed single and multi-modal models. At the end of this work, Chapter 7 summarizes and Chapter 8 systematically addresses the inherent limitations of the studies conducted and outlines the opportunities for future research that arise from these limitations.

2 Related Work

Robot Grasping: The work builds on a long line of work in and related to the field of robot grasping. A significant amount of work in robotics has studied grasping models, which typically make use of manually defined grasping metrics [49, 13, 45]. While these methods enable insight into the physical interactions in grasping, the application of these grasp metrics to real world outcomes is error-prone due to model misspecification. To overcome this issue, some works proposed data based approach to predict grasp outcomes from a wide range of setups like human-supervision [22, 47, 28], simulation [24, 21, 34, 33] or autonomous robotic data collection [41, 29], using visual or depth information. However, these methods are only applicable to a limited extent to reason about direct measurements like contact forces and pressure. To obtain a better estimate of local geometry, some works have used wrist-mounted cameras, which are able to measure the depth of a scene [54]. Additionally, Gualtieri et al. [14] used convolutional neural networks to detect grasping poses from depth data. In this work, simulated over-the-shoulder cameras are used to obtain RGB images. Additionally, these visual images are taken from inside a simulation environment [35] instead of real world.

Touch Sensing: The use of touch sensors for robotic manipulation tasks has become increasingly popular in recent years. Some touch sensors have been proposed in the literature [58] and it turned out that they aid robot grasping in various ways. Due to this observation, low-cost high resolution tactile sensors were developed [27, 39] which are also usable in simulation [56]. In their works, Schill et al. [48] and Bekiroglu et al. [3] used tactile readings to estimate grasp stability. Lie et al. [30] built on this approach and use tactile readings for establishing a grasp adjustment mechanism in case of a low grasp stability prediction. Moreover, several studies [46, 53, 51] have employed tactile readings for slip detection, enabling the application of increased grasping force in situations where an object exhibits signs of slipping from the grasp. Several research studies have introduced robotic systems that incorporate both visual and tactile information for grasping through model-based approaches [1, 36, 19, 15, 2]. These methods have demonstrated enhancements in grasping performance when compared to single-modality inputs. Expanding this research, Calandra et al. [8] showed that integrating vision and touch sensing into a multi-modal framework leads to a performance improvement in prediction of the success rate of a single grasp. Further studies [6, 31] also demonstrated that within a multi-modal framework, the integration of vision and tactile sensing can serve not only to assess grasp stability upon contact establishment but also to acquire grasp learning capabilities.

Machine Learning: A great deal of work has been done in the area of Machine Learning (ML) and in particular Deep Learning (DL) for image classification [23]. Over the years, Convolutional

Neural Networks (CNN) emerged as a prominent approach for image analysis and classification [43, 16]. A number of standard Convolutional Neural Networks have been developed for addressing the challenge of low error image classification, starting with AlexNet [26] and VGGNet [50] up to Residual Neural Networks (ResNet) [17]. Especially the ResNet architecture emerged as a powerful tool since it allows the training of deeper networks by solving the vanishing gradient problem [4]. In this work, a modified ResNet50 architecture is used as substantial part of single and multi-modal neural networks due to the high performance and low error rate for image classification tasks. Prior work successfully applied Convolutional Neural Networks to tactile readings to estimate the material [60, 59]. In these works, weights pretrained on natural image data were used for initializing the Convolutional Neural Networks. Inspired by the success of this approach, the proposed models are initialized with weights pretrained on ImageNet [9] for the ResNet part.

This study draws considerable inspiration from the research conducted by Calandra et al. [8]. Notably, Calandra et al. undertook empirical robot grasping experiments in real-world settings. In contrast, the present study transposes the experimental domain for robot grasping into a simulated environment. While Calandra et al. used a robot gripper for grasping, this work relies on a more complex four-finger Allegro hand with 16 degrees of freedom. Additionally, this work makes use of omnidirectional high-resolution tactile sensors mounted on the fingertips of the multi-finger hand instead of using GelSight tactile sensors [10] for tactile readings. Further, this work proposes a new architecture for processing the tactile readings through single and multi-modal models. A notable departure from the methodology used by Calandra et al. concerns the treatment of sensor gel deformations. While Calandra et al. characterized these deformations as temporal derivatives occurring between object contact and the absence of object contact, this study focuses exclusively on the tactile readings obtained at the precise moment of object contact establishment. Since the multi-finger hand allows the training of models on the combination of tactile readings originating from single fingers, this work draws additional attention to the question of how the performance of tactile models changes with different tactile training data.

3 Background

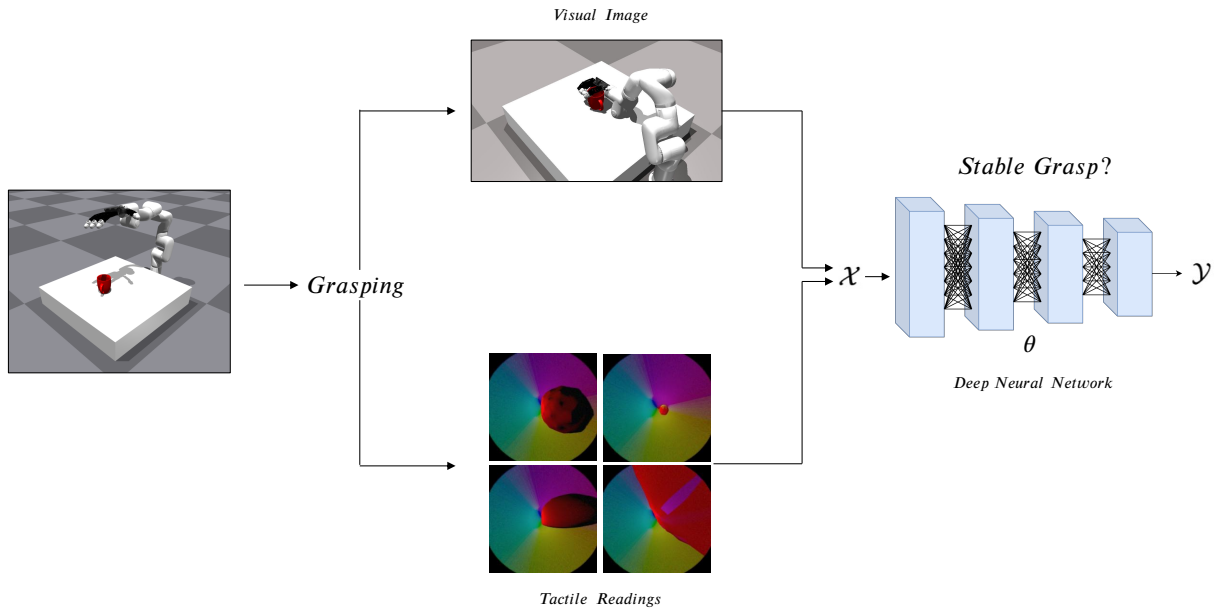


Figure 3.1: In the simulation environment, a robotic system initiates a grasping action on an object. Subsequently, the stability of the grasp is assessed through the utilization of deep neural networks, which process both visual and tactile sensory data acquired during the grasping operation.

The primary goal of this study is to learn a functional mapping $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X}$ represents the domain encompassing a collection of images, while $\mathcal{Y} = [0, 1] \subseteq \mathbb{R}$ denotes a domain encompassing probabilities indicating the likelihood of a successful grasp. It is also often said that the function f_{θ} describes the grasp stability. More formally, the set $\mathcal{X} := \mathcal{P}(X_{vis} \cup X_{tact})$ is defined as the power set of the union of the set of visual X_{vis} and tactile X_{tact} images and $\mathcal{Y}$ is the set of probabilities indicating if a grasp can be considered to be successful given an input sample $s \in \mathcal{X}$. The function f_{θ} is parameterized by the set of weights $\theta \in \Theta$ of a deep neural network. In the process of learning the function f_{θ} , the framework of supervised learning is employed, operating on a labeled dataset denoted as $D = (S, y)$. Here, $S \subseteq \mathcal{X}$ represents a collection of individual samples, while $y : S \rightarrow \{0, 1\}$ is a function that assigns binary labels to each specific sample $s \in S$. Within this context, the objective of learning f_{θ} can be framed as the resolution of a binary classification problem.

The formalism described above is not new [8, 48, 3] and its selection driven by several ad-

vantages and considerations that underlie the methodology of this study. First, the choice of representing the learning task as a functional mapping $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ offers a high degree of interpretability. By mapping images to probabilities within the range $[0, 1]$, it provides a clear and intuitive interpretation of the model's output, which corresponds to the likelihood of a successful grasp. Further, by framing the objective as a binary classification problem, the research focus is sharpened on a fundamental aspect of grasp analysis. This simplification of the problem allows a goal-oriented and comprehensible approach to the learning task, since it is based on the central goal of determining the grasp success. Lastly, the adoption of supervised learning leverages the wealth of information contained in the labeled dataset to train the model effectively. Although this formalism offers clear advantages, one must also be aware of its limitations. One limitation is that it oversimplifies the multiple nuances of grasping scenarios and may neglect more subtle information about the quality of grasping. It also assumes that the binary labels assigned to samples are definitive indicators of success, which may not capture the full range of grasping outcomes. While this formalism is a valuable starting point, future work could explore more sophisticated representations and learning paradigms to address these limitations and improve the accuracy of grasp analysis.

Within the scope of this study, the discourse revolves around the concept of grasping. In this context, a single grasp is defined as an experimental process characterized by the following successive phases: moving the hand to a random position close to the object, then closing the hand randomly, and then attempting to lift the object from its initial position. One can define a grip even more narrowly as the entirety of object positions and the state of the robot joints at the time of the closed hand.

The proposed grasp definition is driven by several advantages. It simplifies the grasping process by breaking it down into discrete, sequential phases. This simplicity can lead to ease of experimentation and reproducibility, making it accessible for testing and development purposes. In addition, the variation in grasping scenarios due to randomized actions can contribute to a more comprehensive data set that can improve the generalizability of the model. However, there are some limitations that should be taken into account. The randomized grasp strategy may lead to inefficient or suboptimal grasp strategies. It may result in unnecessary trial and error, consuming more time and energy than a purposeful, planned approach. Additionally, the approach does not inherently consider object-specific characteristics or adaptability. It treats all grasping scenarios in a similar randomized manner, potentially missing opportunities for optimized grasping based on object properties.

Since robots can be of great use in the real world, the question arises whether the proposed modeling is sufficient to build a system that could be employed for real world grasping. Incorporating randomness into the process of grasping can improve the adaptability and robustness of the robot in real-world scenarios. Additionally, it serves as a valuable concept if no further information about the object or environment of grasping is known. However, this approach may lack realism compared to real-world grasping scenarios, since it does not fully capture the complexity and unpredictability of objects and environments encountered in practice. Further, the grasping experiments are done in simulation, so the gap between simulated random grasping and real-world applications can be challenging to bridge. The same problems occur, if one wants to use the trained models for predicting grasp outcomes with real world robots. In addition, the models were trained exclusively with data resulting from the simulation. Thus, the applicability of the models for real world grasping experiments also depends on the quality of the simulation data. To fully judge the advantages and disadvantages of the proposed grasping approach, it is necessary to look in detail at its implementation inside the simulation. Further, it is important to evaluate the quality of the obtained visual and tactile readings. The comprehensive analyses of the following chapter will provide answers to

some of these questions, allowing a better perspective on the possibilities and limitations of grasping experiments in simulation.

4 Simulation for Multi-Finger Grasping

The field of robotics has undergone a remarkable evolution over the past few decades, with its application in various fields. However, conducting experiments with real robots is often expensive in time and money, which causes impracticality when used for sampling a large amount of data. Recent advancements in computational capabilities have ushered in a paradigm shift, wherein the exploration of robotics in simulation environments is gaining prominence as a superior alternative for a multitude of scientific and practical reasons. To meet the demand for high-performance simulations, researchers and engineers build a couple of simulation environments which are widely used in research and industry [52, 35]. Since this work requires a large amount of visual and tactile data obtained by multiple grasping trials, all grasp experiments are performed in a high-performance simulation. Pursuing this approach will save time and hardware by increasing the number of experiments which benefit in proposing stable deep learning models in a later stage. The purpose of this chapter is to investigate whether, in addition to the time and cost benefits, tactile based grasping of robots in simulation is sufficient to mimic real-world experiments while maintaining the quality of the acquired data. Further, it asks whether the proposed grasping strategy is sufficient for obtaining a diverse collection of grasp data.

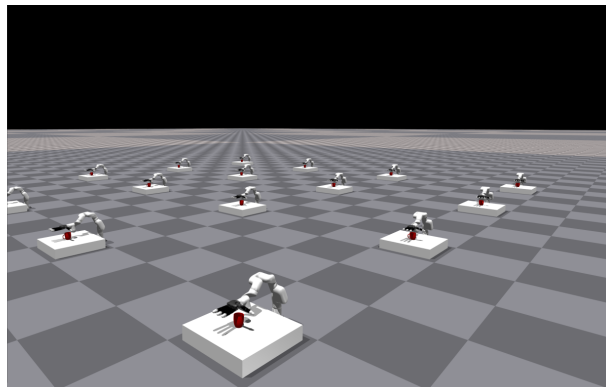


Figure 4.1: Multiple parallel robot grasping experiments in a high performance simulation. The grasp is performed by a four-finger hand mounted at a robot arm. The grasp object (red) is placed on a table in front of the robot arm.

4.1 Preliminaries

For executing grasping experiments, NVIDIA's physics simulation environment Isaac gym [35] in the Preview 4 version is used. The robotic manipulator employed for these experiments is a four-finger Allegro hand equipped with omnidirectional high-resolution tactile sensors, as visually depicted in Figure 4.2. The experiments are done using only a right Allegro hand, which is mounted at a UFactory xArm with seven degrees of freedom to allow a movement inside the environment space. Due to specific software requirements inherent to the simulation environment, all experimental procedures are exclusively conducted on an Ubuntu 20.04 platform, utilizing a NVIDIA graphics card and Python version 3.8. To emulate the tactile sensory feedback, a software solution grounded in the principles of Tacto [56] is employed. Since this work focuses on learning grasp stability for a single object, all experiments are done with a mug of red color that originates from Google Scanned Objects [11]. This particular choice of object holds significance due to its textured surface, which is visible in the tactile data if the sensor has a good point of contact on the mug. In the ensuing discourse, certain terminolo-

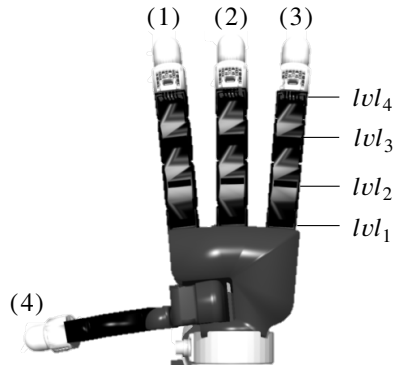


Figure 4.2: Schematic representation of the Allegro hand consisting of Index (1), Middle (2), Ring (3) and Thumb (4). Each joint of a single finger corresponds to a level (*lvl*) ranging from one to four.

gies are employed repeatedly. A grasping experiment entails the comprehensive sequence of motions requisite for executing a singular grasp action, including the lifting of an object. A single environment denotes a cubic spatial region within a broader simulated environment, wherein autonomous experiments are carried out. Each individual environment possesses autonomous control. In instances where the concept of time or time step is invoked, it pertains to the simulation temporal dimension, which can be adjusted in accordance with the particular use case. Whenever the term object is referenced, it exclusively signifies the object in question, namely the mug object. Additionally, it will be referred to joints as a mechanical connection or articulation point that allows two or more rigid components of a robot, typically robot links or segments, to move relative to each other. The number of joints for the hand and the robot arm is equal to their degrees of freedom.

4.2 Robot Control in Simulation

A pivotal component in the execution of a multi-finger hand grasp lies in the precise control of the hand's movements to attain a specific gripping position on the object. The aim of this study is to focus on anticipating the outcome of a particular grasp. Consequently, the initial focus is not on careful planning of the grasping action, but on introducing a controlled degree of randomness into the grasping experiment. Let

$$p_{hand} = [x, y, z] \in \mathbb{R}^3 \quad \text{and} \quad p_{mug} = [\hat{x}, \hat{y}, \hat{z}] \in \mathbb{R}^3$$

denote the position of the palm of the multi-finger hand and the mug at each point of the simulation. Since the palm is the most central point of the hand, its spatial coordinates are referenced whenever talking about poses of the hand. They will also be the main reference when it comes to a basic motion planning. Beside the position, a rotation of hand and mug is specified through a quaternion:

$$q_{hand} = [q_0, q_1, q_2, q_3] \in \mathbb{R}^4 \quad \text{and} \quad q_{mug} = [\hat{q}_0, \hat{q}_1, \hat{q}_2, \hat{q}_3] \in \mathbb{R}^4$$

The initial values of p_{mug} , q_{mug} and q_{hand} are randomized before the start of every new experiment and continuously updated during the simulation. To have a stable base position, the hand can only be rotated around the x- and y-axis. The rotation φ_y around the y-axis is fixed, whereas the rotation φ_x around the x-axis is normally distributed. Therefore, the possible rotations around all axes can be denoted as:

$$\varphi_x \sim \mathcal{N}\left(\frac{\pi}{5}, \frac{\pi}{8}\right), \quad \varphi_y = \frac{\pi}{4}, \quad \varphi_z = 0$$

To avoid an unnaturally strong rotation of the hand, the value of φ_x is clipped to $[0.2, \frac{\pi}{3}] \subseteq \mathbb{R}$. The mug can be placed randomly in a limited area of a single simulation environment. A single environment has a cubic dimension with a side length of 2.0. Since the objects are placed on a cubic table of size $0.8 \times 0.8 \times 0.15$ and the base of the robot arm is grounded at $p_{arm} = [0, 0.5, 0] \in \mathbb{R}^3$ the position of the object is randomized in x and y direction:

$$p_{mug} = [\hat{x}, \hat{y}, \hat{z}] \quad s.t. \quad \hat{x} \sim \mathcal{N}(0.6, 0.005), \hat{y} \sim \mathcal{N}(0.5, 0.005), \hat{z} = 0.15$$

Additionally the object's rotation is determined by a random rotation around the z axis specified by:

$$\hat{\varphi}_z \sim \mathcal{N}\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Before the hand closes, the palm should hover above or lateral to the object depending on the initial angle of the hand. To avoid the occurrence of recurring relative position of the hand to the object, an offset $o = [o_0, o_1, o_2] \in \mathbb{R}^3$ is added to p_{mug} such that:

$$p_{target} = p_{mug} + o = [\hat{x} + o_0, \hat{y} + o_1, \hat{z} + o_2] \in \mathbb{R}^3$$

The offset is calculated by the maximum extension of the mug in every dimension and a randomized normal distributed value for every dimension. In each step, before the physical simulation is executed, a direction vector is calculated

$$v_{dir} = p_{target} - p_{hand} = \begin{bmatrix} \hat{x} - x + o_0 \\ \hat{y} - y + o_1 \\ \hat{z} - z + o_2 \end{bmatrix} \in \mathbb{R}^3$$

which expresses the difference between the current and desired position of the hand. To ensure that the hand always remains at the initial angle, an orientation error is determined by the desired orientation q_{hand} and the current orientation $q_{hand_{curr}}$ at each step, before the physical simulation is executed,

$$q_{err} = q_{hand} * \overline{q_{hand_{curr}}}$$

where $\overline{q_{hand_{curr}}}$ denotes the conjugation of the quaternion. The difference in pose can now be expressed as a vector containing the values of v_{dir} and the three positional coordinates of q_{err} , to which is referred as q_{err} :

$$v_{dpose} = [v_{dir}, q_{err}] \in \mathbb{R}^6$$

To relocate the robotic hand to the target object, only the manipulation of the robot arm is requisite. To achieve this objective, the applied methodology involves the utilization of a jacobian-based inverse kinematics approach. Let $\mathbf{J} \in \mathbb{R}^{6 \times 7}$ denote the jacobian matrix at the hand palm and $\epsilon \in \mathbb{R}^{6 \times 6}$ a damping factor. The seven joints j_{arm} of the robot arm are calculated by

$$j_{arm} = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T + \epsilon)^{-1} v_{dpose}$$

where $\mathbf{J}^T$ denotes the transposed jacobian matrix. The hand is moved towards the mug until it holds:

$$\|v_{dir}\|_F = \sqrt{\sum_{j=1}^3 |v_{dir}(j)|^2} < 0.015$$

When the hand reaches the target position, the first phase of the experiment is completed. The next step described in the following chapter is to close the hand to establish a possible contact with the object.

4.3 Grasping and Evaluation of Success

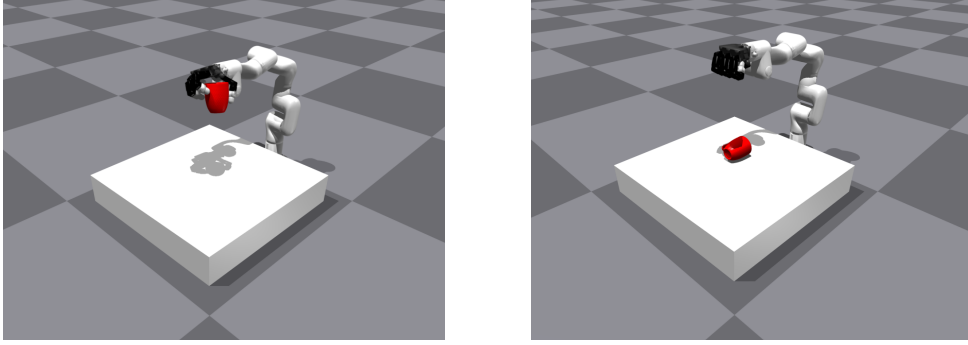


Figure 4.3: Presentation of a successful grasp attempt (left, $l = 1$) and a failed grasp attempt (right, $l = 0$). Whereas in the left picture the robot was able to lift the mug above a predefined threshold, the robot in the right picture failed due to an insufficient grasp configuration.

The complexity of controlling a four-fingered hand arises from the sixteen degrees of freedom of the hand structure. Consequently, closing the hand to perform a grasp requires careful control over each of the sixteen individual joints that make up the hand. In the context of this research, which is primarily aimed at predicting the outcomes of different grasping maneuvers, a degree of randomness is intentionally introduced into the control scheme. This randomness arises from the inherent complexity of precisely regulating all sixteen joints of the hand during the grasping process. The hand consists of four autonomous fingers, the thumb, index, middle, and ring finger. Each of these fingers is equipped with exactly four joints, and the range of values they can assume is restricted by the mechanical limits of the Allegro hand. Let $\mathcal{I}$ denote the set of all joints, $j_{hand} \in \mathbb{R}^{16}$ the joint values and $j_{max} \in \mathbb{R}^{16}$ the maximum values for all joints. To control the closing of the hand, each joint in the same level of the fingers is randomized by a normally distributed value. Let $I \subseteq \mathcal{I}$ denote a collection of joints corresponding to the same level (see Figure 4.2) across all fingers. The values of the joints for grasping are calculated regarding

$$j_{hand}(I) = \lambda_k * j_{max}(I) \quad , \quad \lambda_k \in [0, 1] \subseteq \mathbb{R}$$

while λ_k is normally distributed for each level $k \in \{1, 2, 3, 4\}$ with:

$$\lambda_0 \sim \mathcal{N}(0.7, 0.1), \lambda_1 \sim \mathcal{N}(0.75, 0.1), \lambda_2, \lambda_3 \sim \mathcal{N}(0.85, 0.1)$$

To ensure that the physics simulation has enough time to compute the grasping and the interaction of the fingers and the object, the grasp experiment is blocked for further control actions for a short period of time. After completion of the grasp execution, the robot arm moves the hand to the initial start position. Therefore, the same inverse kinematics method as in 4.2 is used with $p_{target} = p_{init}$ where p_{init} describes the starting position of the hand palm in every single experiment. In particular, it should be mentioned that the rotation of the hand is not changed when returning to the initial start position. If it holds that

$$\|p_{init} - p_{hand}\|_F < 0.05$$

a label $l \in \{0, 1\}$ indicating the grasp success ($l = 1$) or failure ($l = 0$) is calculated. The calculation of the label is done by a function $f : \mathbb{R} \rightarrow \{0, 1\}$ taking as input the $\hat{z}$ -value of p_{mug} and computing:

$$f(\hat{z}) = \begin{cases} 1 & \hat{z} > 0.4 \\ 0 & \text{otherwise} \end{cases}$$

More specifically, a grasp is considered successful if, after the robot arm lifts the object to its initial position, the vertical position of the object exceeds a predefined threshold of 0.4 within the experimental environment. Figure 4.3 shows visual illustrations contrasting successful and unsuccessful grasping events as observed in the simulation.

After the success or failure of a grasp is determined, the experimental environment is systematically reset. This reset process involves repositioning the robot arm to its intended home position, configuring the hand to assume a neutral posture, and then recalculating the positions and orientations of both the hand and the object, as outlined in Section 4.2. After these preparatory steps, a new trial is started.

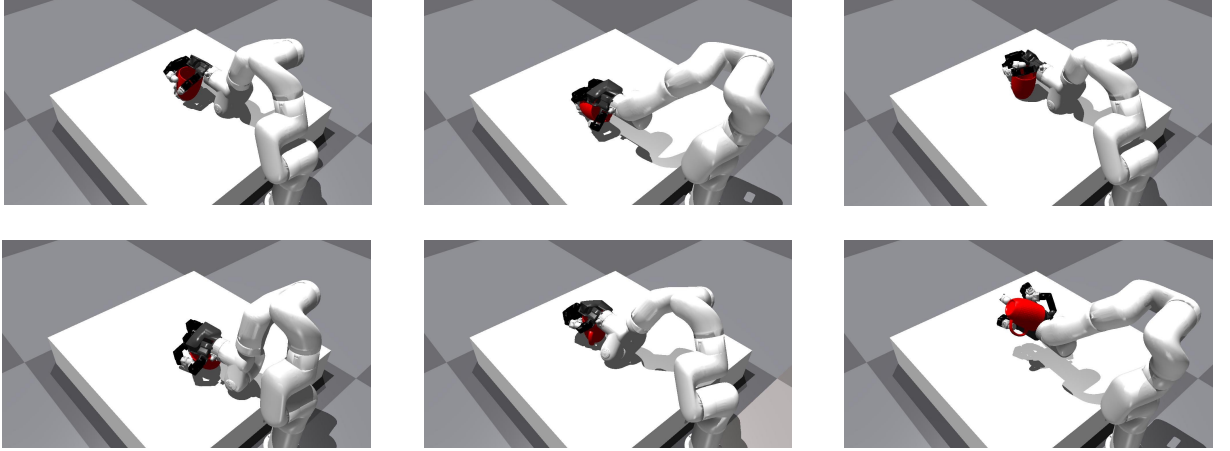


Figure 4.4: Visualization of six different grasp configurations. The different grasp configurations differ in the position and rotation of the hand, the position, and rotation of the mug, as well as how strongly the hand was closed.

4.4 Dealing with Errors

Various simulation anomalies can occur with multiple grasping experiments in a single environment. A major concern is to prevent a single experiment from running indefinitely, which means that it is caught and keeps running. This problem typically occurs in two circumstances. First, when the calculated target position for the hand, denoted p_{target} , cannot be reached accurately due to constraints imposed by the robot's movement capabilities. Secondly, any form of manipulation that results in the cup being moved to a location outside the robot arm's

range of action can cause the object to fall off the table and sometimes even transition to a completely different simulated environment. To mitigate the adverse consequences of this behavior, which can lead to stagnation of the simulation without stopping the experiment, a systematic reset protocol is implemented for each discrete environment. The reset action causes the hand to make an immediate grasping movement and return to its initial position, reviving the experimental context. It is executed whenever a single experiment runs longer than 14 time units. This particular time threshold was figured out to be sufficient higher than the average of execution time for experiments which do not require a reset because of correct termination. Thus, it maintains the probability of a successful grasp, especially when the hand is in close proximity to the target position. It is important to note that an experiment in which a reset occurs is still suitable for further analysis after the reset, since it adheres to the formal sequence of a grasp experiment. To ensure comprehensive monitoring and evaluation of the experiments and their natural termination, it is essential to collect and record both visual and tactile data during the experimental procedure. This comprehensive data sampling serves the dual purpose of allowing continuous review of experimental conditions while providing valuable insight into the dynamic characteristics and outcomes of the experiments.

4.5 Data Collection

The collection of data for robotic tasks plays a crucial role in advancing and fine-tuning robot systems. Especially when it comes to robot learning, the existence of large scale data sets play a crucial role, leading to a range of open sourced data sets for grasping [12] and object manipulation in general [55]. As this work aims to investigate how the use of tactile and visual data advances the prediction of grasp outcome, the collection of data in simulation is a key part for giving an answer.

4.5.1 Preliminaries

In this chapter, the methodology of data acquisition in the simulation environment is explained. Data acquisition is done at four different time stamps T_0, T_1, T_2 and T_3 . T_0 is the time stamp before any robot movement starts. At this stamp, only the visual data is collected. T_1 describes the point where the robot hand has reached the position for executing the grasp. Only the visual feedback is recorded too. At T_2 , the grasp was already executed, and the hand probably has contact with the object. Since T_2 is the most interesting time stamp, the tactile as well as the visual data is collected. The last time stamp T_3 is used to determine the label indicating the grasp success, as explained in 4.3. Just as for T_1 and T_2 , only visual data is saved. If an experiment terminates, all the retrieved data is saved in a single hierarchical data file containing the acquired visual and tactile images at each time step.

4.5.2 Vision

To obtain the visual data, an over-the-shoulder camera is used in every single environment. The camera is placed at $p_{cam} = [-0.3, 1.2, 1.1] \in \mathbb{R}^3$ and viewing to $p_{view} = [0.5, 0.5, 0.15] \in \mathbb{R}^3$ during the whole experiment. Thus, the camera looks exactly at the center of the table on which the grasp object is placed. The visual observation of a camera is updated in every rendering step of the simulation. The camera images are sampled as an RGB image having a width of 1280 and a height of 720 pixels. The horizontal field of view is set to 54.5 degrees for each camera. To ensure the high performance of the simulation, the camera pictures are saved into tensors and updated in place during every rendering step. The tensors are only accessed at the time stamps pointed out in Section 4.5.1. A visual illustration of the collected images depending on time can be seen in Figure 4.5.

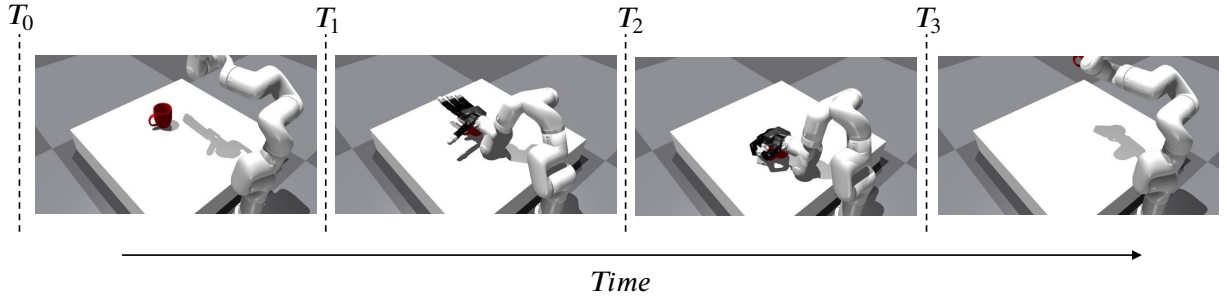


Figure 4.5: The visual data obtained from the simulation environment at each time step. The pictures are obtained from over-shoulder cameras as described in 4.5.2 and show a successful grasp experiment.

4.5.3 Tactile

A major part of this work is the integration of tactile readings into simulated robot grasping, as well as investigating their benefits. Since it is not possible to integrate real-world sensors directly into the simulation environment, this study uses simulation software for the omnidirectional touch sensor based on Tacto [56]. To obtain the readings, the sensors simulator uses the position and orientation of each fingertip as well as the mesh information of the mug to calculate the expected tactile feedback. The tactile reading of each finger is a RGB fish-eye image of size 512×512 pixels with a horizontal field of view set to 220 degrees, a radial distortion of $\xi = 0.3$ and a tangential distortion of $\alpha = 0.3$. The tactile feedback is only calculated at time stamp T_2 , as pointed out in 4.5.1. An illustration of the resulting tactile data when object contact is established can be found in Figure 4.6.

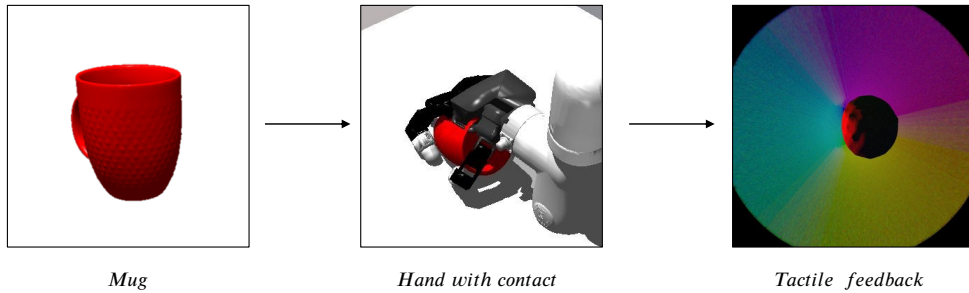


Figure 4.6: Illustration of tactile data collection. The hand tries to grasp the mug. If the grasp configuration is sufficient and direct contact is made between the sensors and the mug, deformations are visible in the tactile readings.

4.6 Evaluation

A major advantage of using simulation environments to conduct experiments with grasping robots is the exceptional computational efficiency they offer. To illustrate this efficiency, more than 56,000 experiments could be performed on a single device within two days. Such a volume of experiments would be difficult for a physical robotic system to handle in a comparable time frame. Furthermore, the simulation framework eliminated the need for additional hardware, as all required sensors and sensor data could be seamlessly integrated and emulated in the simulated environment. These results show that the already expected advantages in terms of experimental duration and cost also apply to the presented approach.

Within the simulation environment, the adoption of advanced grasp planning algorithms to determine an optimal grasp configuration is notably absent. Instead, a substantial number

of parameters governing grasping actions are subjected to controlled randomization to introduce variability. While this stochastic approach facilitates the accumulation of a data set comprising visual and tactile feedback from a wide spectrum of grasp attempts, it is characterized by a notably low rate of success. Specifically, only 425 out of 56,498 attempted grasps resulted in successful object manipulation, yielding an approximate success rate of 0.75 percent. It is noteworthy that a subset of these successful grasps does not exhibit discernible tactile contact in the tactile images. This phenomenon can be attributed to the positioning of omnidirectional sensors exclusively at the fingertips. Notably, the hand's ability to achieve a successful grasp without direct fingertip contact is not unexpected, mirroring instances observed in human grasping scenarios where objects are manipulated utilizing other segments of the fingers. Thus, it can be said that the proposed grasping strategy alone is not sufficient for obtaining a diverse set of data sampled. Especially, the low success rate could prove to be a problem when it comes to model training and thus also for the reliable prediction of positive grasp attempts.

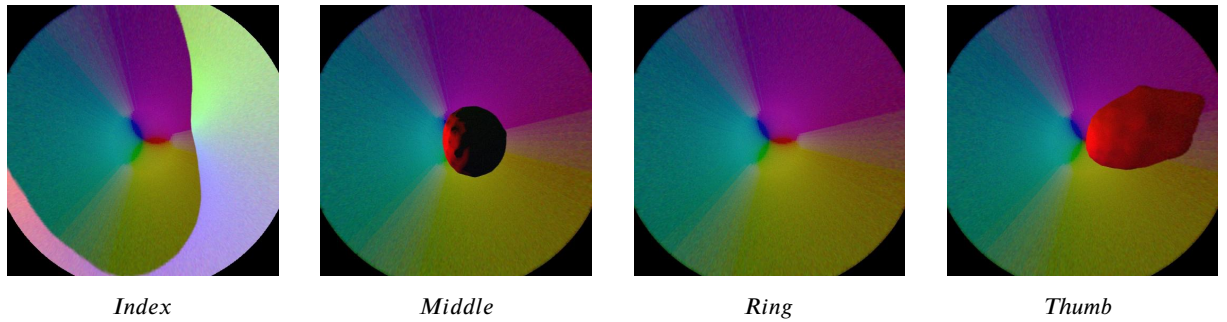


Figure 4.7: The tactile data obtained from the omnidirectional sensor mounted at each fingertip after establishing contact to a mug. Only the Index, Middle, and Thumb have contact with the object, visible through the tactile deflections in the image. The Ring does not have any kind of contact.

However, the simulated tactile readings, when contact is established, demonstrate a good quality as shown in 4.7. They even enable the discernment of textural properties of the object, particularly in areas with discernible surface textures. In tandem with the tactile data, the visual data provisioned by the simulation offers an unobstructed view of both the hand and the object. These visual data meet all required formal specifications, making them suitable for training deep learning models, increasing the utility and potential for comprehensive analysis in an experimental context. Summarizing this, it can be seen that tactile bases grasping with multi-finger hands in simulation is sufficient to collect data with good quality. Nevertheless, it should be noted that there is some space for improvement. Specifically named, the addition of realistic lightning as well as natural noise could help to obtain even more realistic data.

The suitability of the acquired data for effective training of deep learning models can only be determined by specific experimental evaluations. Given the inherent dependence of model performance, especially in the context of supervised learning, on both the quality of the underlying data set and the chosen model architecture, the following chapter explains the architectural details of single-modal and multimodal models.

5 Learning Grasp Stability

In recent years, the field of artificial intelligence has witnessed a remarkable surge in popularity, with machine learning emerging as a front-runner in the technological landscape. One particularly shining star in this area is multi-modal deep learning. Multi-modal deep learning represents a cutting-edge approach that leverages the power of deep neural networks to process and extract valuable insights from multiple data modalities [37]. Humans inherently rely on a multifaceted perceptual approach when engaging with their environment. This innate ability to process and synthesize information from various sensory modalities, such as vision, touch, and proprioception, enables us to navigate, interact, and manipulate objects seamlessly. Drawing inspiration from this fundamental aspect of human cognition, the integration of multi-modal perception has emerged as a pivotal asset in the realm of robotics, particularly in tasks involving grasping and manipulation [38, 57]. By harnessing the synergy between visual data, tactile feedback, and proprioceptive information, robotic systems gain a richer and more comprehensive understanding of their surroundings. This chapter presents how the data from the simulation can be composed into a data set that serves as the foundation for supervised training of single-model and multi-modal models. In addition, the architectures for the single-modal and multi-modal models are presented, and some important design decisions are explained that should enable a high training and test performance of the models.

5.1 Data Preprocessing

As explained in Section 4.5, a total of 56,498 distinct experiments were conducted within the simulation environment. Due to the discrete storage of data for each individual experiment in separate files, the initial step entails merging these discrete datasets into a unified dataset conducive for model training purposes. The merging process involves the extraction of pertinent data elements, namely, the visual information from timestamps T_0 and T_2 , the tactile data from timestamp T_2 , and the corresponding binary label acquired at T_3 . While the label is only one binary integer, the visual and tactile data have a high memory-intensive resolution with 1280×720 and 512×512 pixels. In their work, Richter et al. [44] showed that a too high as well as a too low resolution of input images for Convolutional Neural Networks (CNN) results in a loss of test accuracy. Since a CNN is used as a backbone for the deep-learning models (see Section 5.2), the visual images will be reduced to 320×224 pixels and the tactile images to 224×224 pixels.

Furthermore, an additional dataset is constructed, wherein the tactile images are organized in a square format, each occupying an equivalent area as exemplified in Figure 5.1. To achieve this configuration, the reduced tactile images, originally sized at 224×224 pixels, are employed,

resulting in a final composite image dimension of 448×448 pixels. In alignment with the proposal established for the first dataset, these images are uniformly resized to dimensions of 224×224 pixels. Notably, the accompanying visual data and binary label remain consistent with those in the first dataset. When referring to the tactile information within the second dataset, it will be denoted as the stacked tactile image. It is imperative to ensure the suitability of all data for training deep learning models, and as such, a rigorous screening process is enacted to identify and exclude readings characterized by uniform black or white RGB color values.

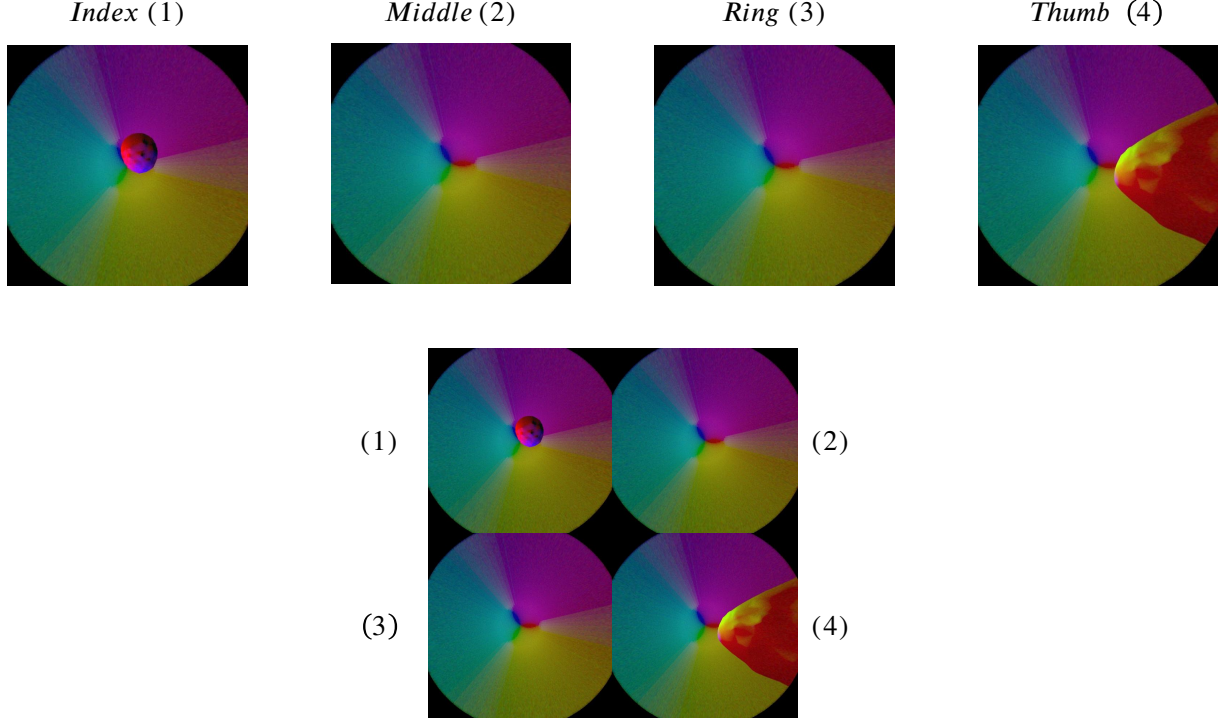


Figure 5.1: A single stacked tactile image consisting of the four tactile readings from the fingertips of the four-finger hand arranged in a square format.

5.2 Deep Learning Models

A central role in answering whether the use of touch is helpful in predicting the outcome of a particular grasp is played by the design of a deep neural network for both single modalities and multiple modalities. The different neural networks are represented by their learnable parameters $\theta \in \Theta$, where Θ denotes the parameter space. Every network θ is designed to compute a function f_θ such that $y = f_\theta(x) \in [0, 1] \subseteq \mathbb{R}$, where y denotes the probability of the grasp being successful and $x \in \mathcal{P}(X_{vis} \cup X_{tact})$ is a set of a single image or of multiple images depending on the proposed model. Due to the proven high performance of Convolutional Neural Networks (CNN) in the area of image classification, a ResNet50 [17] model is used as the backbone for all models.

5.2.1 Single Modality

In the context of single modality models, it is imperative to clarify that these models exclusively receive input from either visual or tactile sources, but not both simultaneously. Formally, it is learned a function f_θ that calculates the probability of a grasp being successful given either

visual $x_{vis} \in \mathcal{P}(X_{vis})$ or tactile $x_{tact} \in \mathcal{P}(X_{tact})$ data:

$$f(\hat{x}) = y \in [0, 1] \subseteq \mathbb{R} \quad \text{for} \quad \hat{x} \in \{x_{vis}, x_{tact}\}$$

The ResNet50 architecture, as introduced in [17], is designed to accept input images characterized by three color channels and dimensions of 224×224 pixels. Given the uniformity in image dimensions across all input samples, the architectural configuration of these models is primarily influenced by the quantity of input images rather than their specific content. To comprehensively assess the performance of single modality models, it is crucial to construct models capable of processing one, two, and four input images, thus enabling an in-depth analysis of their behavior under varying input scenarios. Figure 5.2 shows the architecture of a model

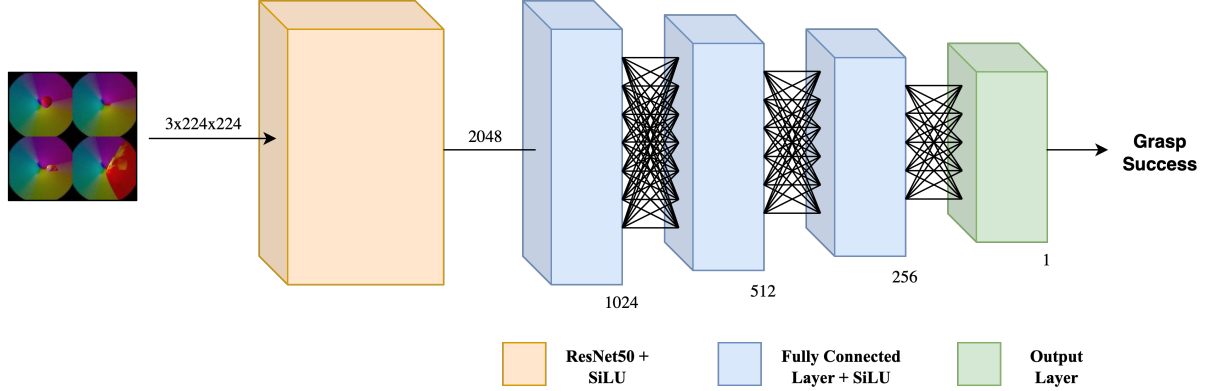


Figure 5.2: Architecture of a single modality model predicting the grasp outcome based on a $3 \times 224 \times 224$ stacked tactile image. Following the initial processing of the image data through a ResNet50 network, the resulting output is subsequently subjected to a series of four fully connected layers, each connected to the preceding layer with SiLU (Sigmoid-weighted Linear Unit) activation functions interposed between them.

processing a single stacked tactile image. The initial stage of processing entails the traversal of the input data through a ResNet50 architecture, with the omission of its final fully connected layer. The resultant output produced by the ResNet is subsequently subjected to activation through the Swish [42] function, also referred to as the Sigmoid-weighted Linear Unit (SiLU) function, which is formally expressed as:

$$SiLU(x) = x * \sigma(x)$$

Here, $\sigma(x)$ represents the mathematical formulation of the logistic sigmoid function. The SiLU function has become more popular over the past years since it tackles the problem of dying Rectified Linear Unit (ReLU) [32]. This phenomenon characterizes the situation in which many neurons activated by ReLU functions output zero values during the course of model training, thus hindering effective learning. A visual comparison between the SiLU and ReLU functions is provided in Figure 5.3. The activated ResNet output is routed through four fully connected layers, with SiLU activation functions applied again after each hidden layer. The final fully connected layer within the network serves as the output layer, responsible for producing a singular numeric output without the application of any activation function. Subsequently, during the validation phase at a later stage, a sigmoid activation function is applied to the output layer's output. This transformation facilitates the conversion of the network's output into a probability value, which in turn signifies the likelihood of a successful grasp. An identical model architecture is employed for variants of the model, which accept a single visual image captured at time

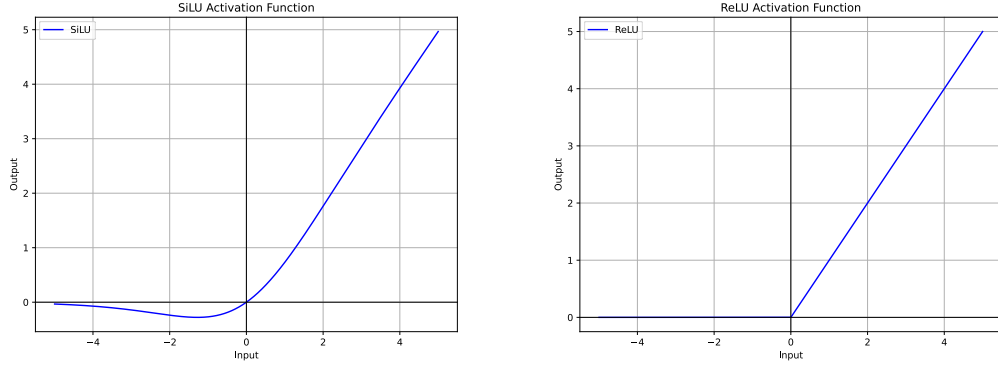


Figure 5.3: Comparison of the SiLU (left) and ReLU (right) activation function. The SiLU function shows improvements over the ReLU function, since it solves the dying ReLU problem by returning values not equal to zeros for input values less than zero.

step T_2 , the tactile readings from the thumb and the tactile readings from the index as its input. However, this model retains the same architectural configuration as depicted in Figure 5.2.

Given the acquisition of visual data at multiple distinct time steps, it is possible to employ more than a single image as input to the model, thereby enhancing the quantity of input data available for grasp outcome prediction. One approach to this end involves the utilization of two images: one captured prior to the proximity of the hand and object (T_0), and the other at the moment the hand already performed the grasp (T_2). Since concurrently processing two input images with separate ResNet networks can be computationally inefficient within a larger neural network architecture, a more streamlined approach is adopted through the concept of shared weights. The shared weights mechanism facilitates the routing of each input image through a common ResNet backbone, employing identical ResNet weights for each image. Consequently, the respective outputs generated by the ResNet for each image are concatenated and subsequently subjected to processing via fully connected layers within the network. As shown in Figure 5.4, the output dimensionality of the ResNet after processing two input

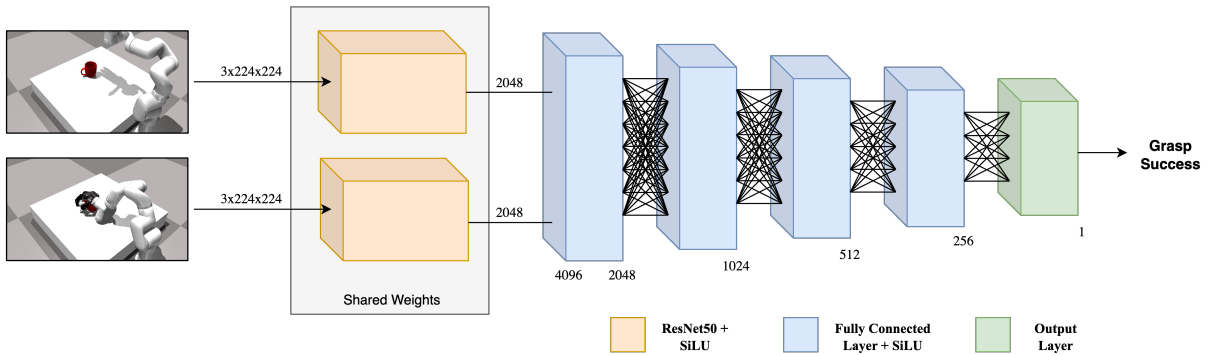


Figure 5.4: Architecture of a single modality model predicting the grasp outcome based on two $3 \times 224 \times 224$ visual images. Following the initial processing of the images through a ResNet50 network using shared weights, the resulting output is concatenated and subsequently subjected to a series of five fully connected layers, each connected to the preceding layer with SiLU activation functions interposed between them.

images is twice that of the configuration shown in Figure 5.2. To account for this increase in dimensionality, an additional fully connected layer is added to the model so that there are a total of five fully linked layers in the architectural structure. However, the overall approach of

the model remains consistent with that of the single image input model, except for the introduction of this additional fully connected layer.

The model designed to accommodate the simultaneous processing of all four tactile readings extends the shared weights concept to a more extensive portion of the model architecture. As visualized in Figure 5.5, the model's backbone is constructed around a ResNet50 network, followed by three fully connected layers. Each of the four input tactile images is independently processed through this network. Notably, unlike the models depicted in Figures 5.2 and 5.4, this model incorporates only three fully connected layers subsequent to the ResNet, yielding individual output vectors of dimensionality 64 for each input image. The outputs generated by the backbone network for each image are concatenated and subsequently passed through two additional fully connected layers. Activation functions are applied using the SiLU function, both between the fully connected layers and after the processing by the ResNet.

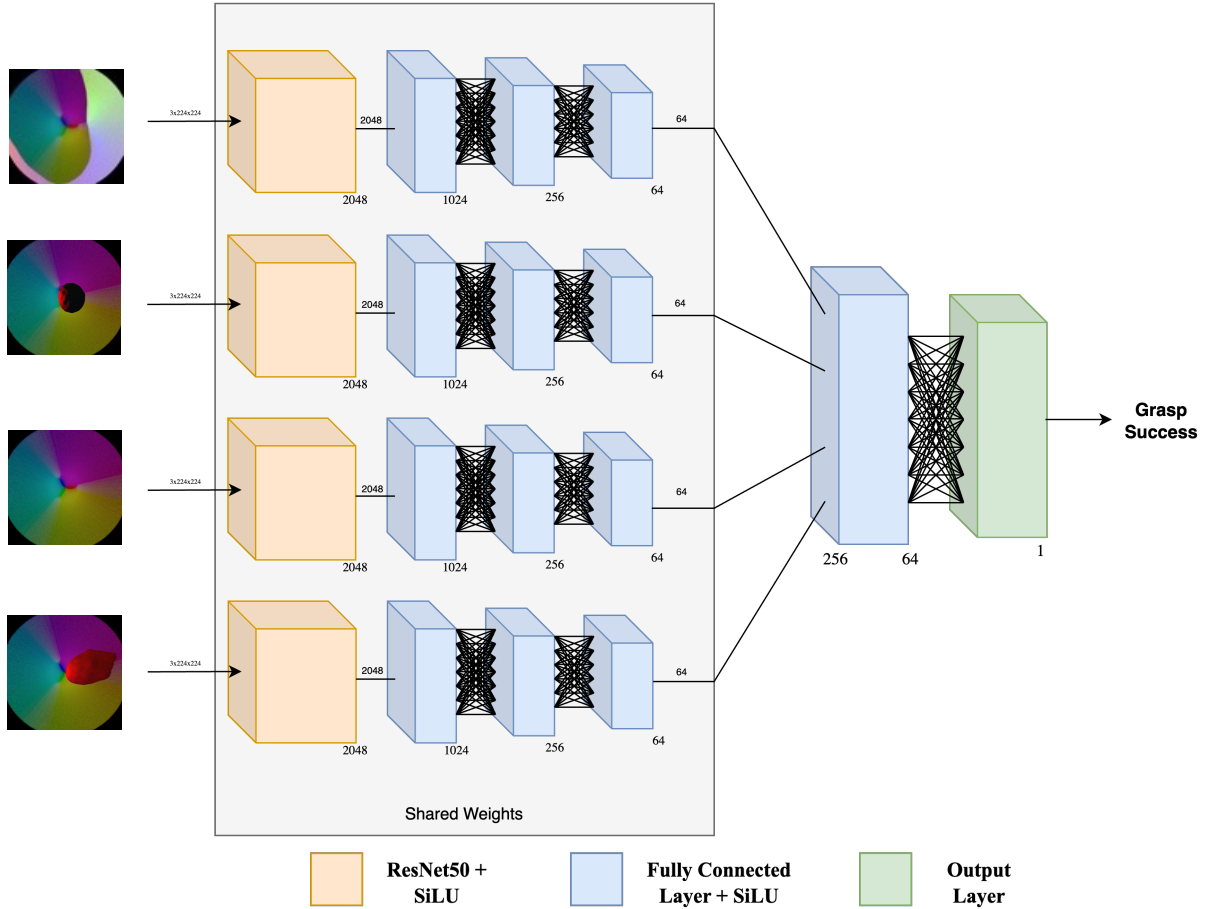


Figure 5.5: Architecture of a single modality model predicting the grasp outcome based on four $3 \times 224 \times 224$ tactile images. Following the initial processing of the images through a ResNet50 network and three fully connected layers using shared weights, the resulting output is concatenated and subsequently subjected to two fully connected layers. Between all fully connected layers, the SiLU activation function is interposed.

5.2.2 Multi Modalities

The multi-modal deep learning models make use of both vision and touch to predict the outcome of a grasp. Formally, it is learned a function f_θ that calculates the probability of a grasp

being successful given both visual $x_{vis} \in \mathcal{P}(X_{vis})$ and tactile $x_{tact} \in \mathcal{P}(X_{tact})$ data:

$$f_{\theta}(x_{vis}, x_{tact}) = y \in [0, 1] \subseteq \mathbb{R}$$

To match the formalism proposed in Chapter 3 the problem can also be formalized into learning a function f_{θ} that calculates the probability of a grasp being successful given a set $\hat{x} \in \mathcal{P}(X_{vis}) \cup \mathcal{P}(X_{tact})$ of both visual and tactile data:

$$f_{\theta}(\hat{x}) = y \in [0, 1] \subseteq \mathbb{R}$$

To produce such a multi-modal model, the models for single modalities are fused. In this study, the intermediate fusion approach is adopted, influenced by the findings of Boulahia et al. [5], which demonstrated superior test results achieved through intermediate fusion when training models with disparate types of images. To construct an intermediate fused multi-modal model, a pivotal modification involves the removal of the output layer of each single modality model, which is then substituted with a fully connected layer generating an output vector of dimensionality 64. It is noteworthy that, in the case of the model dedicated to processing single tactile images, this particular adjustment results in a layer where the input vector possesses an identical dimensionality as the output vector. This contrasts with the behavior observed in all other models, where this stage of the model typically reduces the dimensionality of the input vector. The two output vectors are fused and subsequently subjected to three fully connected

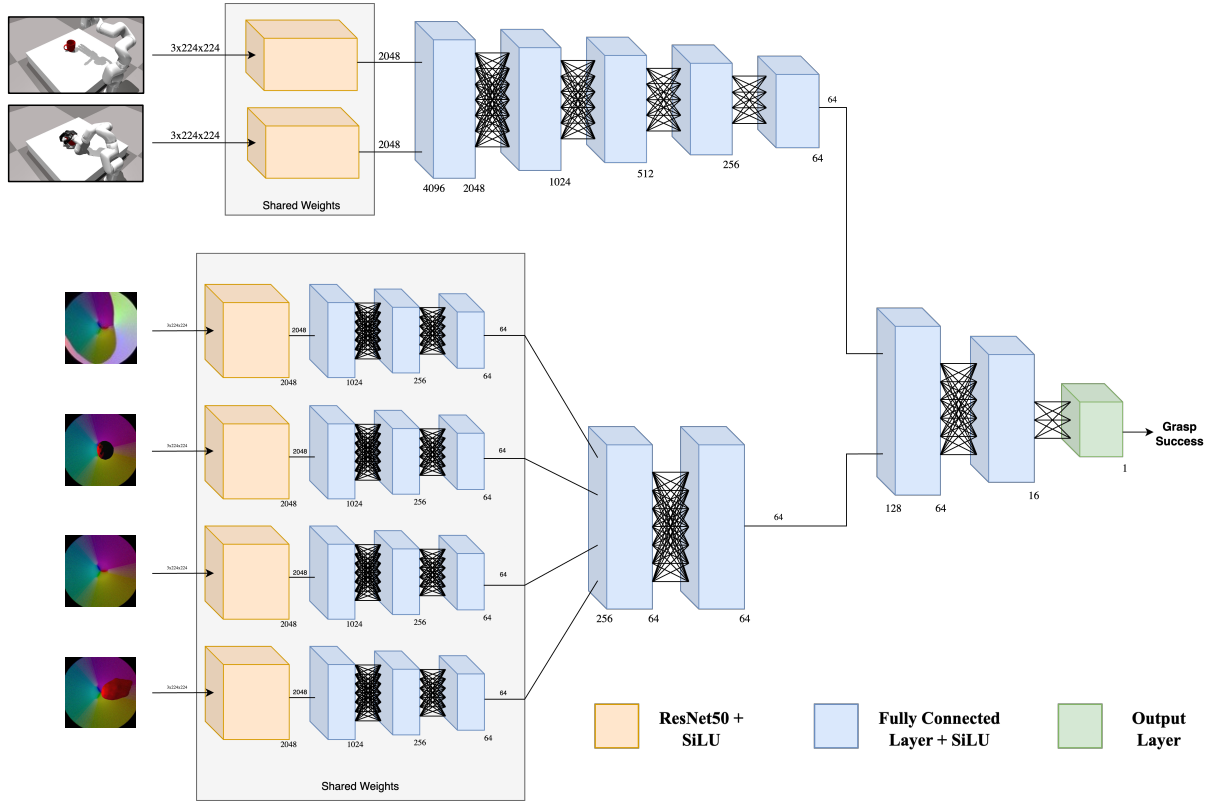


Figure 5.6: Visualization of the architectural configuration of a multi-modal model, which integrates both visual and tactile readings through intermediate fusion. This model accepts a dual input, comprising two $3 \times 224 \times 224$ images for the visual component and four $3 \times 224 \times 224$ images for the tactile component. After merging the individual component results, these consolidated results are further processed through a sequence of three fully connected layers. The SiLU activation function is used both after the fusion process and between the fully connected layers.

layers. Within the architecture of the multi-modal models, the SiLU activation function is applied both between the last three fully connected layers and following the fusion of the modality vectors. As illustrated in Figure 5.6, the multimodal model, designed for two visual and four tactile frames, adheres to this architectural framework. Importantly, all other proposed multi-modal models follow a unified approach. The single modality models, as explained in section 5.2.1, serve as the base models for each modality. Subsequently, these models are fused in an intermediate step characterized by the removal of the output layer, which is replaced by a fully connected layer that produces an output vector of dimensionality 64.

5.2.3 Conclusion

In this chapter, the architectural designs of deep learning models have been presented, which serve as the foundation for the learning of a function denoted as f_θ capable of calculating grasp stability. Emphasis has been placed on design decisions aimed at facilitating an optimal learning outcome. In addition, the methodology for merging data from individual experiments into a labeled data set that forms the basis of the supervised learning problem was explained. Notably, the approach of reorganizing tactile readings into a unified square format has been introduced, offering a novel possibility for tactile data processing.

With the model architectures now established, the investigation of whether the models can be stably trained with the provided data and whether they subsequently yield good test results for unknown data samples becomes of interest. In Chapter 6, answers to these questions will be sought, and a final assessment will be made regarding whether the utilization of tactile readings significantly contributes to the prediction of grasp outcomes.

6 Experiments

Even when employing advanced Deep Learning (DL) architectures, the assessment of a model's proficiency in learning a function can only be determined through the process of training and testing. Chapter 5 presented and discussed the architectures of both single and multi-modal models. In this section, the results of the model training and testing are analyzed to finally conclude whether touch sensing aids in predicting the outcome of a grasp. To comprehensively evaluate models trained with different modalities, an extensive range of different input configurations is tested and evaluated. Vision-only models are trained that include both single-image and dual-image input variations. In the tactile domain, models are trained using either all four tactile frames or exclusively the stacked tactile image. Building on the foundation of the single modality models, a series of multi-modal models will be examined that use different combinations of visual and tactile models. In addition, models using only tactile images of the thumb and index finger will be trained to recognize their role in providing primary stability, similar to human grasping. To investigate the performance of selected individual fingers in combination, two additional models are trained: one employing both thumb and index tactile data and the other using data from the middle and ring fingers. The following sections present and discuss the outcomes of model training and testing. To ensure consistent nomenclature when referencing specific models, the designations provided in Table 6.1 are utilized throughout these sections to identify individual models and their respective input modalities.

Model	Input Data
Vision Single	Visual image at T_2
Vision Double	Visual images at T_0 and T_2
Tactile Stacked	Stacked tactile image
Tactile Single	Four single tactile images
Index	Tactile image from index
Thumb	Tactile image from thumb
Middle + Ring	Tactile images from middle + ring
Index + Thumb	Tactile images from index + thumb
Vision Single + Tactile Stacked	Visual image T_2 , Stacked tactile image
Vision Single + Tactile Single	Visual image T_2 , Four single tactile images
Vision Double + Tactile Stacked	Visual images T_0 and T_2 , Stacked tactile image
Vision Double + Tactile Single	Visual images T_0 and T_2 , Four single tactile images

Table 6.1: Nomenclature of the presented models with overview of the respective input data.

In Section 6.2, the results demonstrate the feasibility of obtaining stable training outcomes using the proposed model architectures and the collected data. Section 6.3 reveals that tactile-only models outperform vision-only models in terms of accuracy. Furthermore, it illustrates that integrating vision and touch into a multi-modal model results in a more balanced model

in the context of precision and recall values. In Section 6.3.3, it is evident that multi-modal models require extended training periods due to their larger size to achieve improved results during training. Moreover, Section 6.3.3 highlights that multi-modal models surpass tactile-only and vision-only models in terms of accuracy when provided with sufficient training time. The section also underscores that increased training duration leads to enhanced balance between accuracy and recall achieved by the multi-modal models, relative to the single-modal models. Further, this chapter closes with a conclusion in Section 6.4.

6.1 Experimental Setup

All models are trained for 25 epochs, utilizing a batch size of 16 and an initial learning rate set to $1e-3$. The Adam [25] optimizer is employed in the optimization process. The ResNet models benefit from weight initialization via pretraining on the ImageNet [9] data set, while all other model weights are initialized randomly using a consistent manual seed. To prevent the model from becoming entrenched on a training plateau, a learning rate scheduler is implemented. This scheduler reduces the learning rate by a factor of 0.1 if neither training accuracy nor loss exhibit improvement over three consecutive epochs. Given that the task of predicting grasp stability can be cast as a binary classification problem, the Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss) function is chosen as the loss criterion. This function seamlessly combines a sigmoid layer with the Binary Cross Entropy (BCE) loss function, providing numerical stability advantages over using a sigmoid function for the output layer followed by the application of the BCE loss function. Thus, the loss l_n for a single sample $n \in \{1, \dots, 16\}$ in a batch of size 16 is calculated as:

$$l_n = y_n * \log(\sigma(x_n)) + (1 - y_n) * \log(1 - \sigma(x_n))$$

where x_n represents the model prediction and y_n denotes the correct label. To assess the significance of the training and validation statistics, a 5-fold cross-validation is performed. Since the dataset is very unbalanced, with only 425 successful compared to 56,073 unsuccessful grasp trials, a stratified 5-fold cross validation procedure is used to have the same ratio of positive and negative examples in each training and test set. Additionally, the training batches are balanced to have a balanced number of positive and negative data examples in each training batch. The batches are not balanced during the test phase. All input images undergo a nor-

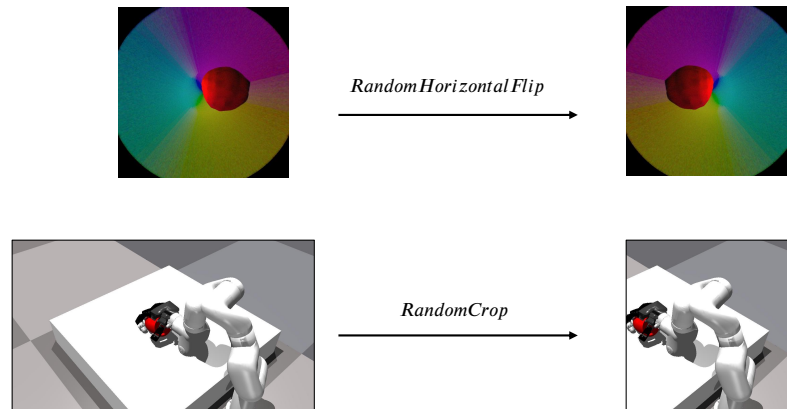


Figure 6.1: Data augmentation techniques employed for the models input images. Horizontal flipping is randomly applied to all input images, shown exemplarily for the tactile image. Moreover, random crops are extracted from the visual images as an additional augmentation strategy, visible in the picture below. These techniques collectively contribute to enhancing the model's generalization capabilities.

malization process with a mean vector of $mean = [0.485, 0.456, 0.406]$ and a standard deviation vector of $std = [0.229, 0.224, 0.225]$. To mitigate the risk of overfitting, data augmentation techniques are introduced. Specifically, visual images, initially sized at 320×224 pixels, undergo random cropping to produce 224×224 pixel images. Furthermore, both tactile images and the randomly cropped visual images are subjected to random horizontal flipping. The data augmentation processes applied to both visual and tactile images are visually depicted in Figure 6.1. The implementation of these techniques is anticipated to enhance accuracy on the test set [40].

6.2 Model Training

The main goal is to find out whether tactile perception contributes to the prediction of grasping results, which requires a rigorous comparison of model performance across different input modalities. To make a meaningful comparison, it is imperative that all models exhibit stable training behavior; otherwise, comparative analysis would not be statistically significant. The following results will illustrate the stability of training across all models, providing a basis for meaningful evaluation of performance on the test set. As described in Section 6.1, all models are trained under identical conditions to ensure fair comparisons.

6.2.1 Vision-Only Models

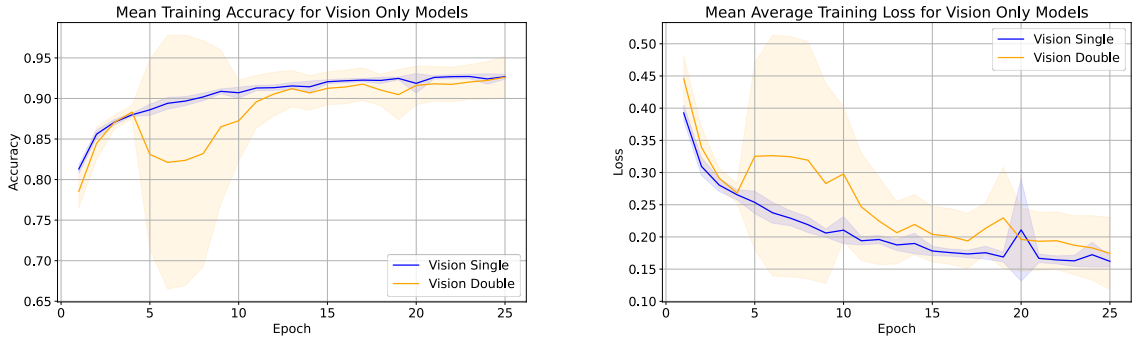


Figure 6.2: Mean and standard derivation for the Average Training Loss (left) and Training Accuracy (right) for the vision-only models trained for 25 epochs on a batch size of 16, applying 5-fold cross validation. While the blue Vision Single curve shows a regular course in accuracy and loss, the orange Vision Double curve shows an irregular course.

Figure 6.2 provides a visual representation of the training dynamics, showing differences between the Vision Single and Vision Double models. The Vision Single model exhibits a more consistent and stable training trajectory, beginning with an accuracy of 81.3% and an average loss of 0.393 during the first epoch. In contrast, the Vision Double model begins training with an accuracy of 78.6% and an average loss of 0.446. While the Vision Single model shows relatively regular curves for average loss and accuracy throughout training, the Vision Double model shows greater irregularities. Particularly, between the fourth and seventh epochs, a divergence in performance becomes evident. During this interval, the accuracy of the Vision Double model declines to 82.1%, while the Vision Single model continues to show improvement. A similar pattern emerges in the curves depicting average loss, with the Vision Double model's loss surging to 0.326 after the fourth epoch, whereas the Vision Single model's loss continues to decrease. Furthermore, a noticeable deflection in the curves of the Vision Single model occurs

at epoch 20, where the accuracy decreases to 91.8%, and the average loss ascends to 0.211. Both models exhibit some minor fluctuations in their curves, but these deviations from the surrounding values are minimal. Upon concluding the 25 epochs, the Vision Single model achieves an accuracy of 92.7%, whereas the Vision Double model attains an accuracy of 92.6%. Despite the close proximity of these accuracy values, a more substantial difference becomes apparent in the final average loss. The Vision Single model concludes with an average loss of 0.162, while the Vision Double model reports a loss of 0.174. In summary, the training statistics indicate a slight advantage in favor of the Vision Single model, characterized by more stable and consistent training behavior in comparison to the Vision Double model.

6.2.2 Tactile-Only Models

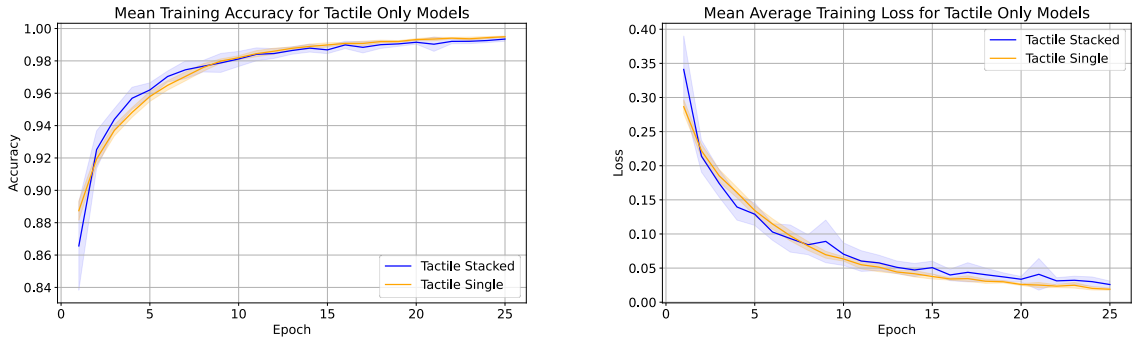


Figure 6.3: Mean and standard derivation for the Average Training Loss (left) and Training Accuracy (right) for the tactile-only models trained for 25 epochs on a batch size of 16, applying 5-fold cross validation. Both models show a stable course in their accuracy and average loss curves. The achieved values of both models after the end of the training are close to each other.

Figure 6.3 reveals the training dynamics of both the Tactile Stacked and Tactile Single models, exhibiting remarkably stable curves in terms of accuracy and loss. This stability is evident through the low values of standard deviation and the infrequent minor fluctuations observed in the respective curves. Analyzing the accuracy curves of the tactile models, the Tactile Single model achieves an accuracy of 88.7%, surpassing the accuracy of the Tactile Stacked model, which attains 86.5% after the same epoch. Notably, in comparison to the initial epoch of the vision-only models, both tactile-only models exhibit higher training accuracy, showcasing an improvement of over 5.0%. After the 25 training epochs, the Tactile Stacked model has an accuracy of 99.3%, whereas the Tactile Single model achieves 99.5%. It's worth noting that the final accuracy of each tactile-only model surpasses the final accuracy of the vision-only models by over 6.0%. Interestingly, the Tactile Stacked model outperforms the Tactile Single model in terms of accuracy from epoch 2 to 8. However, after epoch 8, the Tactile Single model consistently exhibits a higher accuracy than the Tactile Stacked model. A similar pattern emerges in the loss curve. In the initial epoch, the Tactile Stacked model records an average loss of 0.341, whereas the Tactile Single model reports a loss of 0.286 for the same epoch. Notably, the tactile-only models exhibit a lower loss after the first epoch compared to the vision-only models, with a difference exceeding 0.05. After completing 25 training epochs, the Tactile Stacked model concludes with an average loss of 0.026, while the Tactile Single model's loss is slightly lower at 0.019. In comparison to the vision-only models, a substantial improvement is evident, with the best tactile-only model achieving a final average loss that is 0.143 lower than that of the best vision-only model. Additionally, it is noteworthy that the Tactile Single model, which employs four input images, demonstrates more regular curves compared to the Vision Dou-

ble model, which utilizes only two input images. This observation is particularly interesting for subsequent result analysis, given that these models represent larger architectures.

6.2.3 Tactile Modality Models

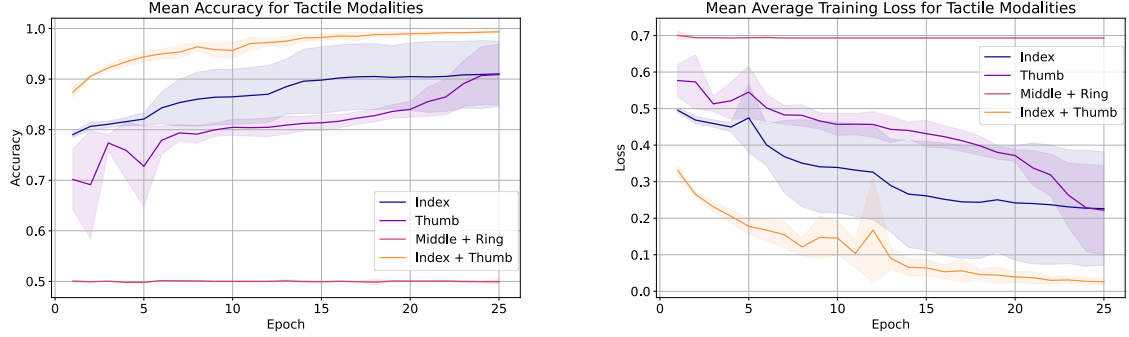


Figure 6.4: Mean and standard deviation for the Average Training Loss (left) and Training Accuracy (right) for the tactile modality models trained for 25 epochs on a batch size of 16 applying 5-fold cross validation. The best scores are achieved by the Index+Thumb model (orange). The Index model (blue) is slightly better than the Thumb model (purple) with a strong convergence at the end. The Middle+Ring (pink) model performs purely in both accuracy and average loss.

Figure 6.4 illustrates the training accuracy and loss curves for models utilizing subsets of tactile readings to predict grasp stability. The Middle+Ring model exhibits notably unfavorable performance, with both accuracy and average loss curves displaying minimal improvement. This model commences with a training accuracy of 50.0% during the initial epochs and concludes with an accuracy of 49.9% after 25 epochs. Though the accuracy experiences a slight deterioration, the average loss displays a marginal improvement, commencing at 0.7 after the first epoch and decreasing to 0.693 at the conclusion of training. These outcomes collectively indicate that the Middle+Ring model underperforms significantly in comparison to other models.

In contrast, the Index+Thumb model demonstrates the best training performance. Commencing with a training accuracy of 87.4%, it progresses steadily to reach an accuracy of 99.3% upon completion of the 25th epoch. The Index+Thumb model's accuracy curve maintains regularity, with only a minor deviation around the tenth epoch. Conversely, the average loss curve exhibits fluctuations between the seventh and 14th epochs, but it still achieves the best initial loss of 0.331, ultimately decreasing to 0.026 by the conclusion of training. Consequently, the Index+Thumb model performs on par with the Tactile Stacked model and slightly below the Tactile Single model in terms of training performance.

The Index model begins with a training accuracy of 79.0%, improving steadily to 91.0% by the conclusion of the last epoch. Notably, the accuracy curve ascends consistently, with a minor dip of 0.06% between the 20th and 21st epochs. Conversely, the average loss curve exhibits an 0.025 increase at the fifth epoch compared to the previous epoch. These statistics position the Index model below the vision and tactile-only models, although it is worth noting that the Index model's training accuracy and average loss are only 1.6% and 0.064 lower, respectively, than the best vision-only model. Compared to the tactile-only models which use all four feedback images, the Index model lags behind, with differences exceeding 8% in accuracy and 0.2 in average loss.

The Thumb model achieves a training accuracy of 90.9% at the end of the final epoch, which is only 0.1% below the accuracy of the Index model at the same point. Nevertheless, the Thumb

model consistently trails the Index model throughout training, notably in the initial epoch, where the Thumb model scores 70.2% in accuracy compared to the Index model's 79.0%. This observation holds true in the average loss curve as well, with the Thumb model beginning with a loss of 0.577, while the Index model starts at 0.495. Interestingly, the Thumb model displays a shift in curve behavior compared to accuracy, recording an average loss of 0.222 at the conclusion of training, surpassing the Index model by 0.004. The Thumb model's performance positions it below the Middle+Ring model but slightly behind the vision-only models, performing notably worse than the tactile-only and Index+Thumb models.

6.2.4 Multi-Modal Models

Figure 6.5 reveals that the combined model with a Single Vision and Tactile Stacked component consistently maintains lower training accuracy compared to the combined model employing a Tactile Single component alongside the same Vision Single part. The model featuring the Tactile Stacked component initiates training with an accuracy of 80.9%, whereas the model incorporating the Tactile Single component commences with an accuracy of 87.1%.

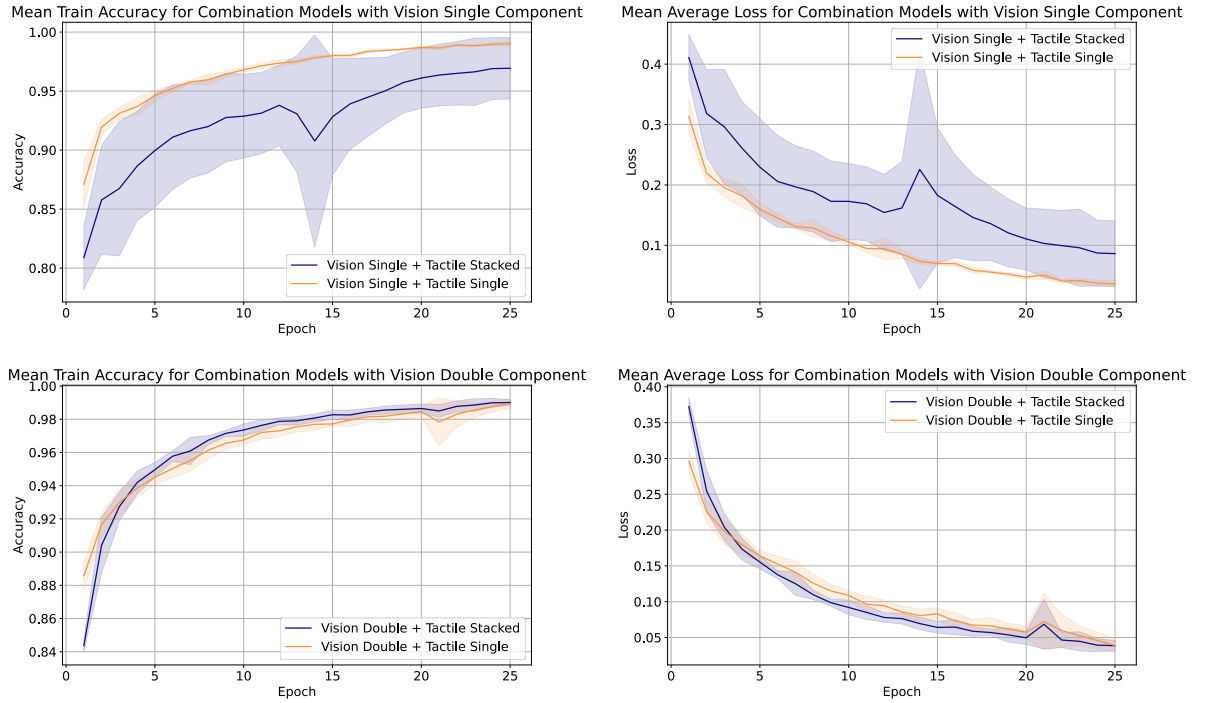


Figure 6.5: Mean and standard deviation for the Average Training Loss (right) and Training Accuracy (left) for the multi-modal models with a Single Vision (top) and Double Vision (down) component trained for 25 epochs on a batch size of 16, applying 5-fold cross validation. The Vision Single + Tactile Stacked model (top, blue) exhibits a notable standard deviation in its training accuracy and average loss. Conversely, all remaining models display consistent learning curves characterized by high accuracy and minimal loss values throughout the entirety of the training epochs.

Throughout the training process depicted in Figure 6.5, the difference in accuracy between the model with the Tactile Stacked component and the model with the Tactile Single component steadily improves. This results in a final accuracy of 96.9% for the model with the Tactile Stacked component and 99.0% for the model with the Tactile Single component. Furthermore, it is evident that the standard deviation of the training accuracy curve is considerably higher for the Vision Single + Tactile Stacked model compared to the Vision Single + Tactile Single model.

This difference is also noticeable when examining the average loss curve for these models. The model utilizing a Tactile Stacked component exhibits a loss of 0.411 after the first epoch, while the model employing a Tactile Single component has a loss of 0.313 during the same epoch. At the conclusion of the training, the loss is improved to 0.086 for the model with a Tactile Stacked component and 0.036 for the model with a Tactile Single component. In light of these observations, it is apparent that the average loss of the model featuring the Tactile Stacked component is more than twice as large as the equivalent value for the model with a Tactile Single component. Additionally, there is a noticeable deviation in both curves of the Vision Single + Tactile Stacked model at epoch 14, where the accuracy decreases from 93.1% to 90.8%, and the loss increases from 0.162 to 0.226 compared to the thirteenth training epoch. Therefore, it can be asserted that the multi-modal models incorporating a Single Vision component exhibit superior training accuracy and loss scores compared to the vision-only models, as well as the Index, Thumb, and Middle + Ring models. However, they still perform slightly worse when compared to the tactile-only models and the Index + Thumb model.

In the lower part of Figure 6.5, it is evident that the curves of the combined models utilizing a Vision Double component appear regular, regardless of the tactile component. Although the training accuracy of the model with a Tactile Stacked component is 84.4% after the initial training epoch, which is lower than the 88.6% achieved by the model with a Tactile Single component, it is noticeable that the accuracy curve of the former model surpasses the latter's curve after the third epoch. This same trend is apparent in the average loss curves on the right side, where the Vision Double + Tactile Stacked model starts with a loss of 0.372, while the Vision Double + Tactile Single model has a loss of 0.296. Upon completing the entire training process, the model with the tactile stacked component attains an accuracy of 99.0%, whereas the other model achieves only 98.95%, demonstrating that the model utilizing a Stacked Tactile component performs better in the end. This finding is further corroborated by the average loss after the final training epoch, which is 0.0386 for the model with a Tactile Stacked component and 0.039 for the model with a Tactile Single component. In comparison to the training statistics of the other models, it is evident that the multi-modal models employing a Vision Double component yield quite similar training results to those using only Vision Single components. In conclusion, these models exhibit better training statistics than the vision-only models, Index, Thumb, and Middle + Ring models, but perform slightly worse than the tactile-only models.

Table 6.2 summarizes the achieved end scores for training accuracy and average loss for the proposed models.

Model	Average Accuracy (%)	Average Loss
Vision Single	92.7 ± 0.3%	0.162 ± 0.009
Vision Double	92.6 ± 2.6%	0.174 ± 0.056
Tactile Stacked	99.3 ± 0.1%	0.026 ± 0.005
Tactile Single	99.5 ± 0.04%	0.019 ± 0.001
Index	91.0 ± 6.5%	0.226 ± 0.155
Thumb	90.9 ± 6.0%	0.222 ± 0.124
Middle + Ring	49.9 ± 0.4%	0.693 ± 0.0
Index + Thumb	99.3 ± 0.2%	0.026 ± 0.009
Vision Single + Tactile Stacked	96.9 ± 2.6%	0.086 ± 0.054
Vision Single + Tactile Single	99.0 ± 0.2%	0.036 ± 0.006
Vision Double + Tactile Stacked	99.0 ± 0.2%	0.038 ± 0.007
Vision Double + Tactile Single	98.9 ± 0.3%	0.039 ± 0.01

Table 6.2: Mean and standard derivation for the Average Training Loss and Training Accuracy for all proposed models trained for 25 epochs and a batch size of 16 applying 5-fold cross validation. The best training accuracy is achieved by the Tactile Single model. The best models in terms of accuracy and average loss for each examined modality are highlighted.

6.2.5 Evaluation

Considering the results discussed in Section 6.2, it becomes evident that the training process for all proposed models consistently exhibits stable accuracy training curves, indicative of reliable and consistent performance during the training iterations. Moreover, the stable loss training curves suggest continuous improvement in model loss throughout the training epochs, signifying a well-behaved training process. These observations collectively indicate that the chosen model architectures do not possess substantial errors hindering learning from the data, and in fact, they yield impressive training statistics when coupled with high-quality training data.

However, an exception is observed in the Middle+Ring model, which displayed no improvement throughout the entire training process. This can be attributed to the nature of the training data, which only sporadically captures tactile feedback from these fingers, resulting in sub-optimal model performance due to the lack of discernible patterns.

Given the strong training performance of the tactile-only model, it is evident that these models can successfully learn patterns from tactile measurements, although they differ from natural images such as those in ImageNet. This observation rules out the possibility that the modeling approach with pretrained weights introduces obstacles that hinder optimal learning. In light of these robust training results, a meaningful comparison of model performance on a test set is warranted to determine whether tactile perception indeed facilitates the prediction of grasp stability for unseen input data.

6.3 Model Testing

The preceding section demonstrated the stability of the training process, establishing a solid foundation for a meaningful performance comparison among the various models. In this section, the fundamental question of the extent to which the inclusion of tactile data contributes to the prediction of grasp stability is addressed. To this end, the models are presented with data samples that have not been encountered during training. It is imperative to note that all models were trained using a stratified 5-fold cross-validation strategy, resulting in a test set containing only approximately 0.75% positive grasp examples. This aspect is important because it implies that relying solely on accuracy values to answer the question may be misleading, as even a model that exclusively classifies instances as the negative class would achieve a high test accuracy. The initial part of the analysis focuses on whether the accuracy values attained by different models provide insights into the utility of tactile data for grasp stability prediction. Subsequently, a deeper examination is conducted, involving scrutiny of the confusion matrix and the application of pertinent metrics to assess the performance of a binary classifier. Furthermore, an investigation is carried out to determine whether the achieved performance is influenced by the number of training epochs. This inquiry helps eliminate the possibility of obtaining erroneous results due to models, particularly larger ones, not having sufficient time to learn intricate patterns.

The test set comprises 11,300 or 11,299 data samples. To maintain consistency with the training process, test samples are presented to the models in unweighted batches of size 16. Importantly, the data augmentation techniques applied during the training phase are not employed on the test samples.

6.3.1 Test Accuracy

Table 6.3 presents the mean classification accuracy on the test set, along with the standard error for all proposed models. Notably, all models exhibit a relatively high accuracy when tested on the validation set. The majority of the proposed models display only minor errors.

6 Experiments

Model	Test Accuracy (%)
Vision Single	97.4 ± 1.6%
Vision Double	89.4 ± 12.3%
Tactile Stacked	98.3 ± 0.5%
Tactile Single	98.1 ± 0.3%
Index	98.1 ± 0.4%
Thumb	96.3 ± 1.8%
Middle + Ring	59.8 ± 54.8%
Index + Thumb	97.9 ± 0.4%
Vision Single + Tactile Stacked	98.0 ± 0.9%
Vision Single + Tactile Single	97.1 ± 1.2%
Vision Double + Tactile Stacked	98.2 ± 0.7%
Vision Double + Tactile Single	97.0 ± 1.8%

Table 6.3: Classification accuracy (report mean and standard error) for models trained with different input modalities. All models are evaluated on a test set with a rate of approximately 0.75% of positive examples. The best test accuracy of 98.3% is achieved by the Tactile Stacked model. The best models in terms of accuracy for each examined modality are highlighted.

However, the Vision Double and Middle + Ring models stand out due to their significantly higher standard errors, which are 12.3% and 53.8%, respectively. These models also happen to exhibit the lowest mean test accuracy among all proposed models.

The highest test accuracy is attained by the Tactile Single model, boasting an impressive score of 98.3% with a minimal error of 0.5%. Slightly trailing behind, the Vision Double + Tactile Stacked model achieves a test accuracy of 98.2% with an error of 0.7%. It is noteworthy that all proposed tactile-only models outperform the vision-only models. Even the best-performing vision-only model achieves a test accuracy of only 97.4%, whereas the tactile-only models consistently surpass the 98.0% mark. Furthermore, the Index and Index+Thumb models outshine the vision models with accuracy scores of 98.1% and 97.9%, respectively.

Additionally, the multi-modal models exhibit test accuracies slightly lower than their single-modal counterparts, but the differences are marginal. For instance, the Vision Double + Tactile Stacked model's mean test accuracy is only 0.1% lower than that of the Tactile Stacked model, which stands as the best-performing single-modal model. Similar trends are observed for other multi-modal models, with differences from the top-performing single-modal component never exceeding 1.1%. It is noteworthy that multi-modal models incorporating a Tactile Single component exhibit slightly lower test accuracy compared to those utilizing a Tactile Stacked component. This observation suggests that larger models with additional input data might require further training for optimal performance.

In summary, based solely on accuracy values, it can be concluded that the use of tactile data contributes to the successful prediction of grasp outcomes. Tactile-only models consistently outperform the vision-only models in terms of accuracy. Except for the Vision Single + Tactile Single model, the multi-modal models consistently outperform their vision-only counterparts. This exception might be attributed to the possibility that the Vision Single + Tactile Single model demands an extended training duration for optimal performance. Furthermore, it is evident that models utilizing tactile information from all fingers consistently outperform or, at the very least, achieve equivalent performance compared to models using only selected finger data. As a result, the subsequent analysis will focus exclusively on the models that leverage information from all fingers.

6.3.2 Confusion Matrix

Due to the imbalanced nature of both the training and test sets, determining the extent to which additional tactile readings aid in predicting grasp stability based solely on model accuracy is challenging. For instance, a model that predicts all instances as non-successful grasps can still achieve an accuracy of over 90.0% on the test set, given the low proportion of positive examples (0.75%) in a dataset comprising 11,300 or 11,299 samples. To mitigate this issue, it is essential to examine the confusion matrices for the various models, with a particular focus on comparing the multi-modal models to their respective single-modal components.

To quantify the samples correctly or incorrectly identified by the model, terms such as True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN) will be employed. To work with relative values and facilitate comparison, the following rates are calculated:

$$\begin{aligned} TPR &= \frac{TP}{TP + FN} & FPR &= \frac{FP}{FP + TN}, \\ FNR &= \frac{FN}{FN + TP} & TNR &= \frac{TN}{TN + FP} \end{aligned}$$

Additionally the metrics of precision (prec) and recall (rec) are calculated:

$$\text{prec} = \frac{TP + TN}{TP + TN + FP} \quad \text{rec} = \frac{TP + FN}{TP + TN + FN}$$

Further the F2-score is calculated to put some more weight on the recall score:

$$F2\text{-score} = 5 * \frac{\text{prec} * \text{rec}}{4 * \text{prec} + \text{rec}}$$

Table 6.4 shows the mean test statistics including the confusion matrix, precision, recall and F2-score for the vision-only, tactile-only and multi-modal models.

Model	TPR	FPR	FNR	TNR	Prec	Rec	F2-Score
Vision Single	70.8%	2.4%	29.2%	97.6%	97.6%	99.8%	99.3%
Vision Double	79.1%	10.5%	20.9%	89.5%	89.6%	99.8%	97.3%
Tactile Stacked	31.1%	1.2%	68.9%	98.8%	98.8%	99.5%	99.3%
Tactile Single	31.8%	1.4%	68.2%	98.6%	98.6%	99.5%	99.3%
Vision Single + Tactile Stacked	60.2%	1.7%	39.8%	98.3%	98.3%	99.7%	99.4%
Vision Single + Tactile Single	49.4%	2.6%	50.5%	97.5%	97.4%	99.6%	99.2%
Vision Double + Tactile Stacked	43.8%	1.4%	56.2%	98.6%	98.6%	99.6%	99.4%
Vision Double + Tactile Single	52.7%	2.6%	47.3%	97.4%	97.4%	99.6%	99.2%

Table 6.4: Mean test statistics including the confusion matrix, precision, recall and F2-score for multiple models trained for 25 epochs and batch size 16. The highlighted multi-modal models show an improvement in their F2-score compared to the single-modal components. The vision-only models achieve the best TPR, whereas the tactile-only models show the best TNR.

Vision-Only Models

The vision-only models exhibit distinct performance metrics. The Vision Double model showcases a higher True Positive Rate (TPR) at 79.1%, while the Vision Single model achieves a TPR of 70.8%. However, the Vision Double model lags behind in the True Negative Rate (TNR) with a value of 89.5%, whereas the Vision Single model excels in TNR at 97.6%. Consequently, the False Positive Rate (FPR) for the Vision Double model is more than double that of the Vision Single model. Furthermore, the Vision Single model displays a greater False Negative Rate

(FNR) at 29.2%, which is 8.3% higher compared to the Vision Double model. Both models attain an identical recall of 99.8%. Notably, there is a substantial difference in precision between the two models, with the Vision Single model achieving 97.6%, while the Vision Double model only reaches 89.5%. This precision gap is also reflected in the F2-scores, with the Vision Single model scoring 99.3%, whereas the Vision Double model achieves a slightly lower score of 97.3%. In comparison to all other proposed models, the vision-only models show the highest TPRs.

Tactile-Only Models

The tactile-only models exhibit strikingly similar performance metrics. Both models achieve a precision score of 99.5% and an F2-score of 99.3%, with only a marginal difference of 0.2% in recall values. Notably, the tactile-only models outperform the vision-only models in precision, albeit with slightly lower recall values. In terms of True Positive Rate (TPR), the Tactile Stacked model achieves 31.1%, while the Tactile Single model reaches 31.8%. However, both models display significantly lower True Negative Rates (TNRs) compared to the vision-only models, with decreases exceeding 35%. This trend is also observed in the False Negative Rates (FNRs), which stand at 68.9% for the Tactile Stacked model and 68.2% for the Tactile Single model. Conversely, the False Positive Rates (FPRs) for the tactile-only models are superior to those of the vision-only models. The Tactile Stacked model slightly outperforms the Tactile Single model in FPR, achieving 1.2% compared to the latter's 1.4%. Similarly, the TNRs for both models are equal at 98.6%. Compared to the vision-only models, the tactile-only models demonstrate notably higher TNR values, exceeding 9% more than the Vision Double model.

Multi-Modal Models

The Vision Single + Tactile Stacked and Vision Double + Tactile Stacked models exhibit the highest F2-scores, both achieving 99.4%, surpassing all other models discussed in this section. The remaining two multi-modal models achieve an F2-score of 99.2%, which is slightly lower than the other proposed models. Except for the Vision Single + Tactile Stacked model, which attains 99.7%, all multi-modal models achieve a recall rate of 99.6%. There is a more noticeable variation in precision values. The Vision Double + Tactile Stacked model demonstrates the highest precision, reaching 98.6%, followed closely by the Vision Single + Tactile Stacked model, which achieves 98.3%. The other two multi-modal models exhibit precision rates of 97.4%. Interestingly, the recall values of the multi-modal models are only slightly lower than those of the vision-only models but slightly higher than those of the tactile-only models. Furthermore, there is an improvement in precision values compared to the vision-only models and a slight deterioration compared to the tactile-only models. An exception to this trend is the Vision Single + Tactile Single model, which does not show an improvement over the vision-only component. The True Positive Rates (TPRs) of the multi-modal models are higher than those of the tactile components but slightly lower compared to the vision component. The Vision Single + Tactile Stacked model achieves the highest TPR at 60.2%, nearly twice as much as the tactile component and only 10.2% less than the vision component. The models using a Vision Double component yield TPRs of 52.7% for the Tactile Single addition and 43.8% for the Tactile Stacked addition. The Vision Single + Tactile Single model attains a TPR of 49.4%. Regarding the False Negative Rates (FNRs), the multi-modal models display lower rates compared to the tactile-only components but higher rates in comparison to the vision-only components. The lowest FNR, at 39.8%, is achieved by the Vision Single + Tactile Stacked model, while the highest FNR of 56.2% is observed for the Vision Double + Tactile Stacked model. The False Positive Rates (FPRs) of the multi-modal models are lower than those of the vision components, except for the Vision Single + Tactile Single model. The improvement in FPR compared to the Vision Dou-

ble model exceeds 7.5%, whereas the improvement relative to the Vision Single model is only 0.7%. This observation suggests an enhancement in True Negative Rates (TNRs) compared to the vision-only components. However, the Vision Single + Tactile Single model again deviates from this pattern. The highest TNR, at 98.6%, is achieved by the Vision Double + Tactile Stacked model.

6.3.3 Influence of Training Time

The multi-modal models use a vision and tactile component, which causes a higher number of parameters that can be learned. Especially since every single modality model has a modified ResNet as a backbone, the number of parameters rises sharply if two single modality models are fused. In the training statistics it could be seen that especially the multi-modal models, which process a high number of input images, lagged behind the smaller ones with their achieved training results.

One possible explanation is that larger models may require more training steps to show their full potential. However, all proposed models are trained for the same number of steps throughout this work. To investigate if the performance of the proposed multi-modal models depends on the number of training steps, these models are retrained for 35 epochs. By analyzing the training and test results, it is possible to give an answer to whether more training leads to better prediction results for the proposed multi-modal models.

Training Statistics

Figure 6.6 presents the mean training accuracy and average loss for the multi-modal models trained for an extended period of 35 epochs. Notably, both accuracy and average loss continue to improve beyond the initial 25 epochs of training. The Vision Single + Tactile Single model attains a final average accuracy of 97.7%, surpassing the 96.9% achieved after 25 epochs. However, it is important to highlight that this model still exhibits a relatively high standard error of 2.6%, in contrast to the other multi-modal models, which all boast accuracy errors below 0.3%. Improvements are also evident in the loss metric, which decreases to 0.062 after 35 epochs, compared to 0.086 after 25 epochs. Although both the Vision Single + Tactile Single and Vision

Model	Average Accuracy (%)	Average Loss
Vision Single + Tactile Stacked	97.7 $\pm$ 2.6%	0.062 $\pm$ 0.054
Vision Single + Tactile Single	99.6 $\pm$ 0.2%	0.017 $\pm$ 0.007
Vision Double + Tactile Stacked	99.7 $\pm$ 0.2%	0.013 $\pm$ 0.007
Vision Double + Tactile Single	99.5 $\pm$ 0.3%	0.021 $\pm$ 0.01

Table 6.5: Mean and standard derivation for the Average Training Loss and Training Accuracy for all proposed models trained for 35 epochs and a batch size of 16 applying 5-fold cross validation. The best training accuracy and loss is achieved by the Tactile Single model. All models demonstrate enhancements in both training accuracy and average loss in contrast to the models trained for 25 epochs. For a detailed comparison, please refer to Table 6.2.

Double + Tactile Stacked models reach a training accuracy of 99.0% after 25 epochs, the latter model exhibits a slight performance boost after 35 epochs, achieving an accuracy of 99.7%. In comparison, the former maintains its accuracy at 99.0% after 35 epochs. Similarly, the average loss for both models sees improvement, decreasing from 0.036 to 0.017 for the Vision Single + Tactile Single model and from 0.038 to 0.013 for the Vision Double + Tactile Stacked model. The Vision Double + Tactile Single model also displays enhanced performance, reaching a higher accuracy of 99.5% after 35 epochs and a lower average loss of 0.021. The ultimate accuracy and

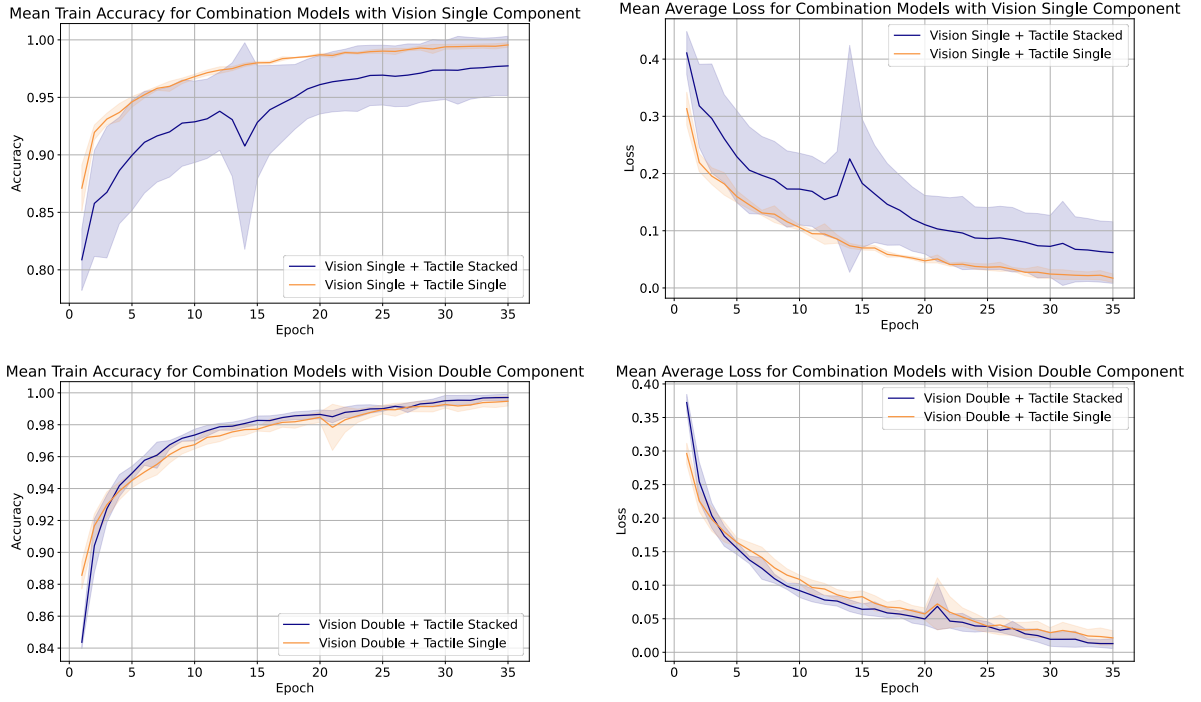


Figure 6.6: Mean and standard deviation for the Average Training Loss (right) and Training Accuracy (left) for the multi-modal models with a single vision (top) and double vision (down) component trained for 35 epochs on a batch size of 16 applying 5-fold cross validation. As already seen in 6.5, the Vision Single + Tactile Stacked model (top, blue) exhibits a notable standard deviation in its training accuracy and average loss. Conversely, all remaining models display consistent learning curves characterized by high accuracy and minimal loss values throughout the entirety of the training epochs.

loss values following 35 epochs of training are presented in 6.5 and can be juxtaposed with the outcomes observed after 25 epochs, as detailed in Table 6.2.

These results demonstrate that prolonged training of the multi-modal models results in increased training accuracy and decreased average training loss. However, it remains essential to investigate whether extended training also enhances the models' ability to predict grasp success for previously unseen examples, as training statistics alone do not provide direct insights into test performance. Further analysis of test performance is necessary.

Test Statistics

Table 6.6 shows that the accuracy of the multi-modal models rises with more training epochs. While the Vision Single + Tactile Stacked model only has a small improvement of 0.1% in comparison to the same model trained for 25 epochs, the Vision Single + Tactile Single model achieved an accuracy of 98.5% which is 1.4% more compared to the same model trained for 25 epochs. The Vision Double + Tactile Stacked model has an accuracy of 98.7% after 35 epochs compared to 98.2 before. The Vision Double + Tactile Single model shows an accuracy of 97.6% improved by 0.6% in comparison to the model trained for 25 epochs. In summary, one can say that more training improves the test accuracy by some percentages. Especially the Vision Single + Tactile Single model has a notable accuracy improvement.

The analysis reveals that extended training epochs contribute to enhanced accuracy in multi-modal models. For instance, the Vision Single + Tactile Stacked model exhibits only a marginal

6 Experiments

Model	Test Accuracy (%)
Vision Single	97.4 ± 1.6%
Vision Double	89.4 ± 12.3%
Tactile Stacked	98.3 ± 0.5%
Tactile Single	98.1 ± 0.3%
Index	98.1 ± 0.4%
Thumb	96.3 ± 1.8%
Middle + Ring	59.8 ± 54.8%
Index + Thumb	97.9 ± 0.4%
Vision Single + Tactile Stacked*	98.1 ± 1.0%
Vision Single + Tactile Single*	98.5 ± 0.4%
Vision Double + Tactile Stacked*	98.7 ± 0.6%
Vision Double + Tactile Single*	97.6 ± 2.2%

Table 6.6: Classification accuracy (report mean and standard error) for models trained with different input modalities. All models are evaluated on a test set with a rate of approximately 0.75% of positive examples. The * marks all models that were trained for 35 epochs. All other models are trained for 25 epochs. The highlighted models illustrate that, with adequate training and input data, the multi-modal model surpasses the single-modal models in terms of test accuracy. Additionally, the tactile-only component outperforms the vision-only component.

0.1% improvement compared to the same model trained for 25 epochs. Conversely, the Vision Single + Tactile Single model attains a notable 1.4% accuracy boost in comparison to its 25 epoch counterpart, achieving an accuracy of 98.5%. Similarly, the Vision Double + Tactile Stacked model achieves an accuracy of 98.7% after 35 epochs, compared to 98.2% previously. The Vision Double + Tactile Single model also demonstrates improved accuracy, reaching 97.6%, an 0.6% increase compared to the model trained for 25 epochs. In summary, it is evident that prolonged training leads to a measurable enhancement in test accuracy, with the Vision Single + Tactile Single model showing particularly noteworthy improvement.

Model	TPR	FPR	FNR	TNR	Prec	Rec	F2-Score
Vision Single	70.8%	2.4%	29.2%	97.6%	97.6%	99.8%	99.3%
Vision Double	79.1%	10.5%	20.9%	89.5%	89.6%	99.8%	97.3%
Tactile Stacked	31.1%	1.2%	68.9%	98.8%	98.8%	99.5%	99.3%
Tactile Single	31.8%	1.4%	68.2%	98.6%	98.6%	99.5%	99.3%
Vision Single + Tactile Stacked*	55.7%	1.6%	44.2%	98.4%	98.4%	99.7%	99.4%
Vision Single + Tactile Single*	33.2%	1.0%	66.8%	99.0%	99.0%	99.5%	99.4%
Vision Double + Tactile Stacked*	27.5%	0.7%	72.5%	99.3%	99.3%	99.5%	99.4%
Vision Double + Tactile Single*	40.0%	2.4%	60.0%	97.6%	97.6%	99.5%	99.1%

Table 6.7: Mean test statistics including the confusion matrix, precision, recall and F2-score for multiple models trained for 35 epochs and batch size 16. The * denotes all models trained for 35 epochs. The highlighted multi-modal models demonstrate an enhancement in their F2-score relative to the single-modal components. Notably, the Vision Single + Tactile Single model improved its F2-score compared to the model trained for 25 epochs, as indicated in Table 6.4. The vision-only models achieve the highest True Positive Rate (TPR), while the tactile-only models exhibit the best True Negative Rate (TNR).

Table 6.7 provides insights into the impact of extended model training on precision, recall, and F2-score. Notably, all models exhibit an improvement in precision values with increased training. The recall value, except for the Vision Single + Tactile Stacked model, which remains constant, experiences a minor 0.1% decrease across all models. Examining the F2-score, it reveals an improvement for the Vision Double + Tactile Stacked model, while a decrement is observed for the Vision Double + Tactile Single model. Moreover, the analysis indicates a

decline in true positive rates (TPRs) and an increase in false negative rates (FNRs) for all models as a consequence of the prolonged training. Specifically, the Vision Double + Tactile Stacked model records the lowest TPR of 27.5%, whereas the highest TPR, at 55.7%, remains consistent with the Vision Single + Tactile Stacked model. Regarding true negative rates (TNRs), the Vision Double + Tactile Stacked model achieves the best performance with a TNR of 99.3%, closely followed by the Vision Single + Tactile Single model, which attains a TNR of 99.0%.

6.3.4 Evaluation of Test Results

It can be seen that the tactile-only models outperform the tactile modality models in terms of accuracy. This behavior is expected since the tactile-only models are trained on all tactile readings, whereas the tactile modalities are trained with a subset of these data sampled. However, it is intriguing to note that the Index and Index of Thumb models achieve accuracy levels closely approaching those of the tactile-only models. This finding reinforces the notion that tactile measurements from the index and thumb fingers tend to be the most influential contributors to grasp stability. Consequently, it appears unnecessary to explore the performance of multi-modal models incorporating a tactile modality component, as it is anticipated that such models would not surpass those using components with access to all tactile readings.

Furthermore, the tactile-only models surpass the vision-only models in terms of accuracy, with a particularly noticeable improvement in contrast to the Vision Double model, which exhibited some instability in its training loss and accuracy curves. An interesting observation emerges when comparing the tactile-only models to the multi-modal models. When all models are trained for an equal number of epochs, the tactile-only models consistently outperform the multi-modal models in terms of accuracy. However, this relationship changes when the multi-modal models undergo additional training, extending by 10 epochs. Given that multi-modal models possess a significantly greater number of learnable parameters, this extension suggests that these models require more training time to achieve optimal performance.

The analysis of the confusion matrix reveals that the vision-only models excel in correctly identifying positive grasps. This proficiency stems from the vision model's ability to recognize favorable hand positions for grasping, even in cases where tactile readings do not explicitly indicate direct contact. This capability proves especially valuable when the hand grasps an object using regions other than the fingertips. However, this strength in positive grasp prediction comes at the cost of reduced accuracy in identifying negative grasps, as indicated by the confusion matrix.

Conversely, the tactile-only models demonstrate remarkable accuracy in predicting true negative grasps. This observation suggests that the tactile-only models are adept at optimizing precision. In summary, it appears that the vision-only models are oriented toward enhancing the recall value, whereas the tactile-only models focus on precision optimization.

The multi-modal models effectively combine these individual strengths. Notably, the recall values of the multi-modal models are slightly lower than those of the vision-only models but higher than those of the tactile-only models. In contrast, the precision values of the multi-modal models are higher than those of the vision-only models but marginally lower than those of the tactile-only models. To quantify this behavior, the F2-score is utilized, providing a metric that establishes a balance between precision and recall, with a slight emphasis on recall.

Especially when considering the results after 35 training epochs, it is evident that the multi-modal models consistently achieve higher F2-scores than the single-component models. This finding underscores that the combination of individual modalities effectively leverages their respective strengths. It is worth noting that an exception to this pattern is observed in the Vision Double + Tactile Single model. This particular model appears to struggle in improving the precision-recall relationship, possibly due to interactions between its single components, especially since the Vision Double model exhibited training and test accuracy weaknesses.

6.4 Conclusion

The initial portion of this section aimed to ascertain the feasibility of training both single and multi-modal models using a dataset generated through simulation. The findings indicated a high degree of training stability for all models, except for one model displaying a relatively high standard error in training accuracy and loss. This robust training stability forms the foundational premise upon which the subsequent comparative analysis of these diverse models is conducted.

The subsequent phase of this section involved a thorough examination of test accuracy across all proposed models, all trained under identical conditions for the same number of epochs. The results revealed that tactile-only models consistently achieved higher accuracy levels than vision-only models, as well as multi-modal models. Furthermore, multi-modal models exhibited superior accuracy compared to vision-only models, strongly suggesting that the integration of tactile sensing noticeably contributes to the prediction of grasp outcomes.

Additionally, a detailed investigation into the confusion matrices and related statistics shed light on the models' individual strengths. Vision-only models exhibited commendable recall capabilities, while tactile-only models consistently demonstrated exceptional accuracy. The F2-score, with a slight emphasis on recall, reinforced the notion that multi-modal models effectively merge the strengths of their constituent modalities, surpassing the individual components in this metric.

The final section of this analysis explored whether extending the training epochs would enhance the multi-modal models' grasp outcome prediction capabilities. The results revealed that increased training epochs led to notable improvements in both training accuracy and loss. Furthermore, this additional training positively impacted test performance, with multi-modal models exhibiting higher accuracy than single components after 35 epochs. This trend was largely consistent, except for one model, when examining the confusion matrices and associated statistics.

In summary, these collective observations underscore the advantageous role of tactile sensing in grasp stability prediction. While tactile readings alone suffice to outperform pure visual approaches in grasp outcome prediction, the fusion of both modalities within an end-to-end multi-modal model yields the most accurate predictions.

7 Summary

This study has demonstrated the beneficial role of tactile sensing in enhancing the predictive capacity of robotic grasp outcomes using a multi-fingered hand in simulation. To this end, a series of end-to-end models, encompassing visual, tactile, and multi-modal configurations, were introduced. A comprehensive examination of these models was conducted, elucidating their respective advantages and drawbacks. These findings yield valuable insights into the efficacy of both single-modal and multi-modal methodologies. Furthermore, it became evident that the multi-modal models excelled in several critical evaluation metrics. This superior performance is attributed to their adept integration of the strengths inherent in both visual and tactile components, underscoring the advantages of such synergistic approaches. This also makes it clear that the multi-modal approach is very promising in the design of deep learning models and must be considered as a serious alternative to classically proven single modality models.

In the initial segment of this study, a randomized grasping system, devoid of advanced grasp planning, was introduced as a means to replicate multi-fingered grasps within a high-performance simulation environment. This approach demonstrated its efficacy in generating a diverse array of grasp scenarios. Nevertheless, it also unveiled a limitation in the form of a scarcity of positive examples within the experimental data.

The experimentation phase encompassed the execution of more than 56,000 individual grasping trials within the simulation framework. Each trial yielded a comprehensive data collection comprising both visual and tactile information, which was subsequently employed to train the single-modality and multi-modal models. It was ascertained that the simulation-generated data provided a robust foundation for model training. However, room for enhancement still exists, especially by incorporating more natural influences such as lighting. Notably, the low success rate observed during the experiments, along with the corresponding data, emerged as a noteworthy constraint in this study.

The findings of this study underscore the significant value of tactile sensing in the domain of robotic manipulation. To enable robots to approach the level of dexterity exhibited by humans in the future, the incorporation of tactile sensing technology should be strongly considered. This technology holds promise not only for application in robotic simulation, but also for integration into real-world robotic systems to further advance the capabilities of robotic manipulation.

8 Future Work

Although this work shows that tactile sensing benefits in robot grasping with multi-finger hands in simulation, there are a few limitations that should be addressed in future work.

Simulation: Conducting robot grasping experiments in simulation save a notable amount of time and money. Nevertheless, it is important to realize that you cannot compare a simulation environment with real conditions. Since the goal should be to use the models trained on simulation data also for real world grasping experiments, the question occurs how the simulation could be improved to become more realistic. A first step in this direction could be the addition of different natural lighting scenarios as well as disturbing influences. Additionally, the control of the robot and especially of the hand has a great influence on the success of a single grasp attempt. Since humans are able to adjust factors such as movement speed and grasp force depending on the grasped object, it would be a major enhancement to provide advanced control schemes of the simulated robot to eliminate scenarios in which the robot cannot grasp objects because the movements are too fast and the force applied is too great.

Machine Learning: The proposed models are trained in a supervised manner on data obtained by a simulation approach. Due to the randomization of grasp configurations, the dataset is very unbalanced, containing less than one percent of successful grasps. To prevent the models from overfitting, data augmentation as well as weighted training batches have been applied. However, the test set is still very unbalanced and the high accuracy, precision and recall values do not provide an insight how these models would perform on a balanced test set of diverse grasps. This raises the question whether the models still show a high performance when facing a diverse, balanced test set. In addition to that, it would be interesting to see how a diverse balanced train set will affect the learning curves of the proposed models. Tackling this questions will also give insights whether the chosen model architectures are capable of predicting grasp outcomes when trained on different datasets.

Robot Grasping: To clarify whether touch sensing aids in learning grasp stability, this work proposes models that are able to predict the outcome of a given grasp. While this provides valuable guidance after a grasp is performed and contact established, it does not help in selecting a successful grasp configuration. Even if the grasp configuration is adjusted if the predicted grasp stability is below a predefined threshold, there is no guarantee that a suitable grasp configuration can be found in a reasonable amount of time. To tackle this issue, one could use an advanced grasp planning based on visual and depth information. However, these approaches do not include touch sensing. Since we showed the importance of tactile sensing and multi modality for simulated robot grasping, the question arises how touch sensing could efficiently

be integrated into grasp planning to select promising grasp configurations before determining the grasp stability. While this work shows the applicability of supervised learning for grasp outcome prediction, it might also be beneficial to investigate whether reinforcement learning aids in predicting promising grasp configurations.

Touch Sensing and Control: The experiments carried out involve the collection of tactile readings at a specific time stamp. While this methodology permits the prediction of grasp stability before the object is physically lifted, it does not offer insights into grasp stability during the lifting phase. Nonetheless, the latter aspect is of significant importance as it provides crucial information for determining whether an object should be lifted, particularly in relation to a specific grasp configuration. The implementation of a closed-loop control system during the lifting of an object also affords the opportunity to dynamically adjust finger poses in response to potential object slippage, thereby mitigating the risk of the object falling out of the hand.

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